Lecture 3, Appendix: Topology, Differential Topology and Functional Analysis

Math for Machine Learning

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Introduction

In Lecture 3, we used several facts without proof: that a continuous function on a closed and bounded set attains its maximum, that a function with continuous partial derivatives is differentiable (Theorem 3.6), the implicit function theorem in the proof of the Lagrange multiplier theorem (Theorem 12.1), and the existence and uniqueness of solutions of differential equations (Theorem 15.2). We also met sets that are not open subsets of $\mathbb{R}^n$, but curved objects: spheres, the set of orthogonal matrices, and level sets of functions. And in Lectures 2 and 4, we worked with vector spaces of functions and of random variables, which are infinite-dimensional. This appendix provides the mathematical background.

- Sections A.1–A.4 are about topology, the mathematics of nearness and continuity. Its central notions are open sets, compactness and connectedness. Topology studies the properties of sets that do not change under continuous deformations.
- Sections A.5–A.9 are about differential topology, which studies manifolds, i.e., sets that look like $\mathbb{R}^k$ near each of their points, and differentiable maps between them. Its main tools are the inverse and implicit function theorems. At the end, Morse theory connects the critical points of a function, classified by the eigenvalues of its Hessian as in Lecture 3, with the shape of the space on which it is defined.
- Section A.10 collects some places where these ideas appear in machine learning.
- Sections A.11–A.14 are about functional analysis, i.e., linear algebra and analysis in infinite-dimensional vector spaces, such as spaces of functions. Its central notions are completeness, orthogonal projections and the spectral theorem for compact operators. They prove the existence theorem for differential equations of Lecture 3 and the Radon–Nikodym theorem of Lecture 4, and they lead to Fourier series, to expansions of random functions, and to the theory of kernel methods. Section A.15 returns to machine learning.

We use the notation of Lecture 3: vectors in $\mathbb{R}^n$ are bold, and $\|\cdot\|$ is the Euclidean norm unless we say otherwise. Elements of general vector spaces, such as functions, are written in normal font.

A.1 Metric Spaces and Open Sets

Definition A.1.1 (Metric space)**.** A metric on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

A set with a metric is a metric space.

Example A.1.2. 1. Every norm on a vector space defines a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|$ (Lecture 2). In particular, $\mathbb{R}^n$ with the ℓ_1, ℓ_2 or ℓ_∞ norm is a metric space.

2. On the space $C[a,b]$ of continuous functions on $[a,b]$, $d(f,g) = \max_{x\in[a,b]} |f(x) - g(x)|$ is a metric.

3. Every subset of a metric space is a metric space with the same metric. For example, the unit sphere $S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$ is a metric space with the Euclidean distance.

4. On any set, the discrete metric $d(x,y) = 1$ for $x \neq y$ (and $d(x,x) = 0$) is a metric.

Definition A.1.3 (Open and closed sets)**.** Let (X,d) be a metric space. The open ball with center x and radius $r > 0$ is $B(x,r) = \{y \in X : d(x,y) < r\}$. A set $U \subseteq X$ is open if for every $x \in U$ there is an $r > 0$ with $B(x,r) \subseteq U$. A set $C \subseteq X$ is closed if its complement $X \setminus C$ is open.

For $X = \mathbb{R}^n$, this is the definition of open sets in Lecture 3. Closed is not the opposite of open: $\mathbb{R}^n$ and $\emptyset$ are both open and closed, and the half-open interval $[0,1)$ is neither open nor closed in $\mathbb{R}$.

Proposition A.1.4. *In a metric space X:*

1. *every open ball is open;*

2. *every union of open sets is open, and every intersection of finitely many open sets is open;*

3. *every intersection of closed sets is closed, and every union of finitely many closed sets is closed.*

Proof. (1) If $y \in B(x,r)$, let $s = r - d(x,y) > 0$. For $z \in B(y,s)$, the triangle inequality gives $d(x,z) \leq d(x,y) + d(y,z) < d(x,y) + s = r$, so $B(y,s) \subseteq B(x,r)$.

(2) If x lies in a union of open sets, it lies in one of them, U, and some ball $B(x,r) \subseteq U$ lies in the union. If $x \in U_1 \cap \cdots \cap U_m$ with $B(x,r_i) \subseteq U_i$, then $B(x, \min_i r_i)$ lies in the intersection.

(3) Take complements in (2). □

Infinite intersections of open sets need not be open: $\bigcap_{k\geq 1}(-\frac{1}{k}, \frac{1}{k}) = \{0\}$. For a set $A \subseteq X$, the interior $\operatorname{int} A$ is the union of all open sets contained in A (the largest open subset of A), the closure $\overline{A}$ is the intersection of all closed sets containing A (the smallest closed set containing A), and the boundary is $\partial A = \overline{A} \setminus \operatorname{int} A$. For example, the closed ball $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$ in $\mathbb{R}^n$ has the open ball as interior and the sphere S^{n-1} as boundary.

Definition A.1.5 (Convergence)**.** A sequence (x_k) in a metric space converges to x if $d(x_k, x) \to 0$ as $k \to \infty$. It is a Cauchy sequence if for every $\varepsilon > 0$ there is an N with $d(x_k, x_l) < \varepsilon$ for all $k, l \geq N$.

Closed sets are exactly the sets that cannot be left by taking limits.

Proposition A.1.6. *A subset C of a metric space X is closed if and only if the limit of every convergent sequence of points of C lies in C.*

Proof. Let C be closed, and let $x_k \in C$ converge to x. If $x \notin C$, then some ball $B(x,r)$ lies in the open set $X \setminus C$; but $x_k \in B(x,r)$ for large k, a contradiction. Conversely, suppose that C is not closed. Then $X \setminus C$ is not open, so there is an $x \notin C$ such that every ball $B(x, \frac{1}{k})$ contains a point $x_k \in C$. Then $x_k \to x$, and $x \notin C$. □

A.2 Continuity and Topological Spaces

Continuity can be described with open sets alone, without mentioning distances.

Proposition A.2.1. *Let $f : X \to Y$ be a map between metric spaces. Then f is continuous (at every point, in the ε-δ sense of Lecture 3) if and only if for every open set $V \subseteq Y$, the preimage $f^{-1}(V) = \{x \in X : f(x) \in V\}$ is open.*

Proof. Let f be continuous, $V \subseteq Y$ open, and $x \in f^{-1}(V)$. There is an $\varepsilon > 0$ with $B(f(x), \varepsilon) \subseteq V$, and by continuity a $\delta > 0$ such that $d(x, x') < \delta$ implies $d(f(x), f(x')) < \varepsilon$, i.e., $B(x, \delta) \subseteq f^{-1}(V)$. Conversely, if preimages of open sets are open, then for $x \in X$ and $\varepsilon > 0$, the set $f^{-1}(B(f(x), \varepsilon))$ is open and contains x, so it contains a ball $B(x, \delta)$; this is the ε-δ condition. □

Taking complements, f is continuous if and only if preimages of closed sets are closed.

Corollary A.2.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and $c \in \mathbb{R}$. Then the set $\{f < c\}$ is open, and the sublevel set $\{f \leq c\}$ and the level set $\{f = c\}$ are closed. The same holds for intersections of finitely many such sets.*

Proof. These are the preimages of the open set $(-\infty, c)$ and the closed sets $(-\infty, c]$ and $\{c\}$. □

So the feasible sets of the constrained problems in Lecture 3, such as $\{\mathbf{x} : g_1(\mathbf{x}) \leq 0, \ldots, g_m(\mathbf{x}) \leq 0\}$ for continuous g_i, are closed. Similarly, the set $\mathrm{GL}(n, \mathbb{R})$ of invertible $n \times n$ matrices is open in $\mathbb{R}^{n \times n}$, as the preimage of the open set $\mathbb{R} \setminus \{0\}$ under the continuous function det (a polynomial in the entries).

Proposition A.2.1 suggests an abstraction: forget the metric, and only remember which sets are open.

Definition A.2.3 (Topological space)**.** A topology on a set X is a collection $\mathcal{T}$ of subsets of X, called open sets, such that $\emptyset, X \in \mathcal{T}$, every union of sets in $\mathcal{T}$ lies in $\mathcal{T}$, and every intersection of finitely many sets in $\mathcal{T}$ lies in $\mathcal{T}$. A map $f : X \to Y$ between topological spaces is continuous if preimages of open sets are open. A homeomorphism is a continuous bijection whose inverse is also continuous; if one exists, X and Y are homeomorphic.

By Proposition A.1.4, the open sets of a metric space form a topology. Two metrics can give the same topology: we will see in Theorem A.3.8 that all norms on $\mathbb{R}^n$ do. Homeomorphic spaces are the same from the point of view of topology: a homeomorphism matches up their open sets, convergent sequences, continuous functions, and so on.

Example A.2.4.

1. The sigmoid function is a homeomorphism from $\mathbb{R}$ to $(0, 1)$, with the continuous inverse $p \mapsto \log \frac{p}{1-p}$ (the logit). Similarly, tanh is a homeomorphism $\mathbb{R} \to (-1, 1)$. So bounded and unbounded sets can be homeomorphic: lengths and distances are not topological properties.

2. The open ball $B(\mathbf{0}, 1) \subseteq \mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$, via $\mathbf{x} \mapsto \frac{\mathbf{x}}{1-\|\mathbf{x}\|}$, with the inverse $\mathbf{y} \mapsto \frac{\mathbf{y}}{1+\|\mathbf{y}\|}$.

3. A square and a disk in $\mathbb{R}^2$ are homeomorphic (rescale each ray from the center), and so are a circle and the boundary of a square: corners are not a topological property. But a circle and an interval are not homeomorphic (Example A.4.3).

A.3 Compactness

The real numbers have no gaps. This is the completeness axiom, which we take as one of the defining properties of $\mathbb{R}$: every nonempty set of real numbers that is bounded above has a least upper bound (supremum). It implies that every bounded monotone sequence of real numbers converges (to the supremum or the infimum of its values).

Theorem A.3.1 (Bolzano–Weierstrass)**.** *Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.*

Proof. First let $n = 1$. Call an index k a peak if $x_k \geq x_l$ for all $l > k$. If there are infinitely many peaks $k_1 < k_2 < \cdots$, then $x_{k_1} \geq x_{k_2} \geq \cdots$ is a monotone subsequence. Otherwise, there is an index k_1 after which there are no peaks; then k_1 is not a peak, so $x_{k_2} > x_{k_1}$ for some $k_2 > k_1$; k_2 is not a peak either, and so on. This gives an increasing subsequence. In both cases, the monotone subsequence is bounded, so it converges. For $n > 1$, choose a subsequence whose first coordinates converge, then a subsequence of it whose second coordinates converge, and so on; after n steps all coordinates converge. □

Corollary A.3.2. *$\mathbb{R}^n$ is complete: every Cauchy sequence in $\mathbb{R}^n$ converges.*

Proof. A Cauchy sequence $(\mathbf{x}_k)$ is bounded (all terms after some N lie within distance 1 of $\mathbf{x}_N$), so by Theorem A.3.1 a subsequence converges to some $\mathbf{x}$. For $\varepsilon > 0$, choose N such that $\|\mathbf{x}_k - \mathbf{x}_l\| < \varepsilon$ for $k, l \geq N$, and an element $\mathbf{x}_l$ of the subsequence with $l \geq N$ and $\|\mathbf{x}_l - \mathbf{x}\| < \varepsilon$. Then $\|\mathbf{x}_k - \mathbf{x}\| < 2\varepsilon$ for all $k \geq N$. □

Definition A.3.3 (Compact set)**.** A subset K of a metric space is compact if every sequence in K has a subsequence that converges to a point of K.

(In general topological spaces, one defines compactness with open covers: K is compact if every collection of open sets that covers K contains finitely many sets that already cover K. For metric spaces, the two definitions agree; we do not prove this.)

Theorem A.3.4 (Heine–Borel)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. Let K be closed and bounded, and let $(\mathbf{x}_k)$ be a sequence in K. By Theorem A.3.1, a subsequence converges, and by Proposition A.1.6, its limit lies in K. Conversely, if K is unbounded, there are $\mathbf{x}_k \in K$ with $\|\mathbf{x}_k\| \geq k$, and no subsequence converges, since convergent sequences are bounded. If K is not closed, there is a sequence in K that converges to a point outside K (Proposition A.1.6), and all its subsequences converge to the same point, so none converges to a point of K. □

For example, spheres, closed balls (for any norm), and the set $\mathrm{O}(n)$ of orthogonal matrices are compact. $\mathrm{O}(n)$ is closed, as the preimage of $\{I\}$ under the continuous map $Q \mapsto Q^\top Q$, and bounded, since $\|Q\|_F^2 = \operatorname{tr}(Q^\top Q) = n$. The hyperbola $\{xy = 1\}$ is closed but unbounded, and the open ball is bounded but not closed; neither is compact.

Theorem A.3.5 (Extreme value theorem)**.** *Let K be a nonempty compact set, and let $f : K \to \mathbb{R}$ be continuous. Then f attains a minimum and a maximum on K.*

Proof. Let $m = \inf\{f(\mathbf{x}) : \mathbf{x} \in K\}$, where we allow $m = -\infty$. Choose $\mathbf{x}_k \in K$ with $f(\mathbf{x}_k) \to m$. A subsequence converges to some $\mathbf{x}^* \in K$, and by continuity, $f(\mathbf{x}^*) = m$. So m is finite, and it is the minimum. For the maximum, apply this to $-f$. □

This justifies the claims in Lecture 3 that $f(x, y) = x + y$ and the Rayleigh quotient $\mathbf{x}^\top A\mathbf{x}$ attain their maxima on the unit sphere (Examples 12.2 and 12.3 of Lecture 3). On $\mathbb{R}^n$, which is not compact, a minimum need not exist, but the following condition is often enough.

Proposition A.3.6 (Coercive functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and coercive, i.e., $f(\mathbf{x}) \to \infty$ as $\|\mathbf{x}\| \to \infty$. Then f has a global minimum.*

Proof. Fix any $\mathbf{x}_0$. The sublevel set $S = \{f \le f(\mathbf{x}_0)\}$ is closed (Corollary A.2.2) and bounded (by coercivity), so it is compact and nonempty, and f has a minimum $\mathbf{x}^*$ on S. Points outside S have values larger than $f(\mathbf{x}_0) \ge f(\mathbf{x}^*)$, so $\mathbf{x}^*$ is a global minimum. □

Example A.3.7. A loss $L(\mathbf{w}) \ge 0$ plus a ridge penalty, $L(\mathbf{w}) + \lambda\|\mathbf{w}\|^2$ with $\lambda > 0$, is coercive, so it has a global minimum whenever L is continuous. Without regularization, a minimum need not exist: for logistic regression with the single training example $x = 1$, $y = 1$, the loss $\log(1 + e^{-w})$ decreases towards 0 as $w \to \infty$, but never reaches it. The same happens for every linearly separable data set, where scaling up a separating weight vector always decreases the loss.

Compactness also explains why all norms on $\mathbb{R}^n$ lead to the same notions of convergence and continuity, as claimed in Lecture 3.

Theorem A.3.8 (Equivalence of norms)**.** *For every norm N on $\mathbb{R}^n$, there are constants $0 < c \le C$ with $c\|\mathbf{x}\| \le N(\mathbf{x}) \le C\|\mathbf{x}\|$ for all $\mathbf{x}$. Consequently, all norms on $\mathbb{R}^n$ define the same open sets.*

Proof. Writing $\mathbf{x} = \sum_i x_i \mathbf{e}_i$, the triangle inequality and the Cauchy–Schwarz inequality give

$$N(\mathbf{x}) \le \sum_i |x_i| N(\mathbf{e}_i) \le \Big(\sum_i N(\mathbf{e}_i)^2 \Big)^{1/2} \|\mathbf{x}\| = C\|\mathbf{x}\|.$$

In particular, $|N(\mathbf{x}) - N(\mathbf{y})| \le N(\mathbf{x} - \mathbf{y}) \le C\|\mathbf{x} - \mathbf{y}\|$, so N is continuous. By Theorem A.3.5, N attains a minimum c on the compact unit sphere S^{n-1}, and $c > 0$ since N is positive on S^{n-1}. For $\mathbf{x} \ne \mathbf{0}$, $N(\mathbf{x}) = \|\mathbf{x}\|\, N(\frac{\mathbf{x}}{\|\mathbf{x}\|}) \ge c\|\mathbf{x}\|$. Finally, the inequalities show that every N-ball around a point contains a Euclidean ball around it and vice versa, so the open sets are the same. □

Remark A.3.9 (Infinite dimensions). In infinite-dimensional spaces, closed and bounded sets need not be compact, and different norms can give different topologies. For example, in the space ℓ^2 of square-summable sequences (Section A.11), the unit sequences $e_1, e_2, \ldots$ lie in the closed unit ball, but $\|e_j - e_k\| = \sqrt{2}$ for $j \ne k$, so no subsequence converges. This is one reason why analysis in function spaces, such as the spaces of functions in Lecture 2, is harder than in $\mathbb{R}^n$ (Sections A.11–A.14).

A.4 Connectedness

Definition A.4.1 (Connected sets)**.** A subset X of a metric space is disconnected if $X \subseteq U \cup V$ for two open sets U and V such that $X \cap U$ and $X \cap V$ are nonempty and disjoint. Otherwise, X is connected. It is path-connected if any two points $x, y \in X$ can be joined by a path, i.e., a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Proposition A.4.2. *1. Every interval of $\mathbb{R}$ is connected.*

2. If X is connected and f is continuous on X, then $f(X)$ is connected.

3. Every path-connected set is connected.

Proof. (1) Suppose an interval I is covered by open sets U and V as in the definition, with $a \in I \cap U$ and $b \in I \cap V$; by symmetry, we may assume $a < b$. Then $[a, b] \subseteq I$. Let $s = \sup\{t \in [a, b] : [a, t] \subseteq U\}$; then $[a, t] \subseteq U$ for every $t < s$, and $s \in [a, b] \subseteq I$, so s lies in U or in V. If $s \in V$, then since V is open, $(s - \varepsilon, s] \subseteq V$ for some $\varepsilon > 0$; but points of $(s - \varepsilon, s)$ also lie in U, which contradicts $I \cap U \cap V = \emptyset$. If $s \in U$, then $s \neq b$ (since $b \in V$), so $s < b$, and since U is open, $[s, s + \varepsilon] \subseteq U$ for some small $\varepsilon > 0$ with $s + \varepsilon \leq b$. Then $[a, s + \varepsilon] \subseteq U$, which contradicts the definition of s.

(2) If $f(X)$ were covered by open sets U and V as in the definition, then the preimages $f^{-1}(U)$ and $f^{-1}(V)$ would disconnect X (they are open by continuity, in the sense that they are of the form $X \cap U'$ for open sets U').

(3) Suppose that open sets U, V disconnect a path-connected set X, with $x \in U$ and $y \in V$. Then the preimages of U and V under a path from x to y would disconnect the interval $[0, 1]$, which contradicts (1). □

Together, (1) and (2) give the intermediate value theorem: a continuous function $f : [a, b] \to \mathbb{R}$ takes every value between $f(a)$ and $f(b)$, since its image is a connected subset of $\mathbb{R}$, i.e., an interval.

Example A.4.3.

1. $\mathbb{R}^n$, balls, and the sphere S^{n-1} for $n \geq 2$ are path-connected, hence connected.
2. The hyperbola $\{xy = 1\}$ is not connected: it lies in the union of the open half-planes $\{x > 0\}$ and $\{x < 0\}$, and meets both.
3. $\mathrm{GL}(n, \mathbb{R})$ is not connected: it is the union of the disjoint open sets $\{\det > 0\}$ and $\{\det < 0\}$. In particular, an invertible matrix with $\det < 0$ (a reflection) cannot be continuously deformed into the identity through invertible matrices. Similarly, $\mathrm{O}(n)$ consists of the two pieces $\det Q = 1$ (the rotations, Lecture 2) and $\det Q = -1$; one can show that the rotations form a connected set.
4. A circle and an interval are not homeomorphic. If we remove a point from a circle, the rest is still connected (it is an arc); if we remove an interior point from an interval, the rest is disconnected. A homeomorphism would map a point and its complement to a point and its complement, and it preserves connectedness by Proposition A.4.2(2).

A.5 The Inverse Function Theorem

From now on, we combine topology with derivatives. A map is smooth if all its partial derivatives of all orders exist and are continuous, and a diffeomorphism is a smooth bijection with a smooth inverse. We first fill a gap from Lecture 3, using a vector-valued version of the mean value theorem.

Lemma A.5.1 (Mean value inequality)**.** *Let $\mathbf{g} : U \to \mathbb{R}^m$ be continuously differentiable on an open set $U \subseteq \mathbb{R}^n$ that contains the segment between $\mathbf{x}$ and $\mathbf{y}$. Then*

$$\|\mathbf{g}(\mathbf{y}) - \mathbf{g}(\mathbf{x})\| \leq \sup_{\mathbf{z} \in [\mathbf{x}, \mathbf{y}]} \|D\mathbf{g}(\mathbf{z})\|_2 \, \|\mathbf{y} - \mathbf{x}\|,$$

where $\|\cdot\|_2$ is the spectral norm of Lecture 2.

Proof. Let $\mathbf{u} = \mathbf{g}(\mathbf{y}) - \mathbf{g}(\mathbf{x})$, and assume $\mathbf{u} \neq \mathbf{0}$. The function $\psi(t) = \mathbf{u}^\top \mathbf{g}(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ has $\psi(1) - \psi(0) = \|\mathbf{u}\|^2$. By the mean value theorem and the chain rule, $\psi(1) - \psi(0) = \psi'(\tau) = \mathbf{u}^\top D\mathbf{g}(\mathbf{z})(\mathbf{y} - \mathbf{x})$ with $\mathbf{z} = \mathbf{x} + \tau(\mathbf{y} - \mathbf{x})$. By the Cauchy–Schwarz inequality, $\|\mathbf{u}\|^2 \leq \|\mathbf{u}\| \, \|D\mathbf{g}(\mathbf{z})\|_2 \, \|\mathbf{y} - \mathbf{x}\|$; divide by $\|\mathbf{u}\|$. □

Proposition A.5.2 (Theorem 3.6 of Lecture 3)**.** *If all partial derivatives of $f : U \to \mathbb{R}$ exist in a ball around $\mathbf{x}$ and are continuous at $\mathbf{x}$, then f is differentiable at $\mathbf{x}$.*

Proof. For small $\mathbf{h}$, let $\mathbf{z}_0 = \mathbf{x}$ and $\mathbf{z}_j = \mathbf{x} + h_1\mathbf{e}_1 + \cdots + h_j\mathbf{e}_j$, so that $\mathbf{z}_n = \mathbf{x} + \mathbf{h}$. The points $\mathbf{z}_{j-1}$ and $\mathbf{z}_j$ differ only in the j-th coordinate, so by the mean value theorem in one variable, $f(\mathbf{z}_j) - f(\mathbf{z}_{j-1}) = \frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j)h_j$ for a point $\boldsymbol{\xi}_j$ between them. Adding these equations,

$$\left|f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h}\right| = \Big|\sum_{j=1}^{n} \Big(\frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j) - \frac{\partial f}{\partial x_j}(\mathbf{x})\Big)h_j\Big| \le \sum_{j=1}^{n} \Big|\frac{\partial f}{\partial x_j}(\boldsymbol{\xi}_j) - \frac{\partial f}{\partial x_j}(\mathbf{x})\Big| \, \|\mathbf{h}\|.$$

As $\mathbf{h} \to \mathbf{0}$, all $\boldsymbol{\xi}_j \to \mathbf{x}$, so the sum tends to 0 by the continuity of the partial derivatives at $\mathbf{x}$. □

The key tool of this section is a simple but powerful theorem about solving equations by iteration.

Theorem A.5.3 (Banach fixed-point theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be closed and nonempty, and let $T : C \to C$ be a contraction, i.e., $\|T(\mathbf{x}) - T(\mathbf{y})\| \le q\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in C$, with a constant $q < 1$. Then T has exactly one fixed point $\mathbf{x}^* = T(\mathbf{x}^*)$. For every $\mathbf{x}_0 \in C$, the iterates $\mathbf{x}_{k+1} = T(\mathbf{x}_k)$ converge to $\mathbf{x}^*$, and*

$$\|\mathbf{x}_k - \mathbf{x}^*\| \le \frac{q^k}{1-q}\|\mathbf{x}_1 - \mathbf{x}_0\|.$$

Proof. By induction, $\|\mathbf{x}_{j+1} - \mathbf{x}_j\| \le q^j\|\mathbf{x}_1 - \mathbf{x}_0\|$. For $l > k$, the triangle inequality and the geometric series give

$$\|\mathbf{x}_l - \mathbf{x}_k\| \le \sum_{j=k}^{l-1} q^j\|\mathbf{x}_1 - \mathbf{x}_0\| \le \frac{q^k}{1-q}\|\mathbf{x}_1 - \mathbf{x}_0\|.$$

So $(\mathbf{x}_k)$ is a Cauchy sequence, which converges to some $\mathbf{x}^*$ by Corollary A.3.2, and $\mathbf{x}^* \in C$ since C is closed. Since T is continuous, $T(\mathbf{x}^*) = \lim T(\mathbf{x}_k) = \lim \mathbf{x}_{k+1} = \mathbf{x}^*$. Letting $l \to \infty$ in the inequality gives the error bound. If $\mathbf{y}^*$ is another fixed point, then $\|\mathbf{x}^* - \mathbf{y}^*\| = \|T(\mathbf{x}^*) - T(\mathbf{y}^*)\| \le q\|\mathbf{x}^* - \mathbf{y}^*\|$, so $\mathbf{x}^* = \mathbf{y}^*$. □

Example A.5.4. The function $T(x) = \cos x$ maps $[0, 1]$ into $[\cos 1, 1] \subseteq [0, 1]$, and $|T'(x)| = |\sin x| \le \sin 1 \approx 0.841$ there, so it is a contraction with $q = \sin 1$ (by the mean value theorem). So the equation $\cos x = x$ has exactly one solution in $[0, 1]$, $x^* = 0.739085\ldots$, and pressing the cosine key of a calculator repeatedly finds it. Starting at $x_0 = 1$, the error bound guarantees an accuracy of 10^{-6} after 87 steps; in fact, 32 steps suffice. Gradient descent on a quadratic function with $\rho < 1$ (Theorem 10.6 of Lecture 3) is another example of a contraction.

Corollary A.5.5 (Invertible residual layers)**.** *Let $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^n$ satisfy $\|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{y})\| \le q\|\mathbf{x} - \mathbf{y}\|$ with $q < 1$. Then the residual layer $\mathbf{f}(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ is a homeomorphism of $\mathbb{R}^n$, and $\mathbf{f}^{-1}(\mathbf{y})$ is the limit of the iteration $\mathbf{x}_{k+1} = \mathbf{y} - \mathbf{g}(\mathbf{x}_k)$.*

Proof. For fixed $\mathbf{y}$, the map $T(\mathbf{x}) = \mathbf{y} - \mathbf{g}(\mathbf{x})$ is a contraction of $\mathbb{R}^n$, and $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ if and only if $\mathbf{x} = T(\mathbf{x})$. By Theorem A.5.3, there is exactly one solution, so $\mathbf{f}$ is bijective. It is continuous, and so is its inverse: $\|\mathbf{x} - \mathbf{x}'\| \le \|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| + \|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{x}')\| \le \|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| + q\|\mathbf{x} - \mathbf{x}'\|$, so $\|\mathbf{f}^{-1}(\mathbf{y}) - \mathbf{f}^{-1}(\mathbf{y}')\| \le \frac{1}{1-q}\|\mathbf{y} - \mathbf{y}'\|$. □

Invertible neural networks used in normalizing flows (Remark 4.8 of Lecture 3) are built from such layers: constraining the Lipschitz constant of each residual branch (e.g., by normalizing the spectral norms of its weight matrices) makes the whole network invertible. A diffeomorphism must have invertible Jacobians everywhere, by the chain rule applied to $\mathbf{f}^{-1} \circ \mathbf{f} = \mathrm{id}$. The inverse function theorem is a local converse.

Theorem A.5.6 (Inverse function theorem)**.** *Let $\mathbf{f} : U \to \mathbb{R}^n$ be continuously differentiable on an open set $U \subseteq \mathbb{R}^n$, and let $D\mathbf{f}(\mathbf{a})$ be invertible. Then there are open sets $V \ni \mathbf{a}$ and $W \ni \mathbf{f}(\mathbf{a})$ such that $\mathbf{f}$ maps V bijectively onto W, and the inverse $\mathbf{f}^{-1} : W \to V$ is continuously differentiable, with*

$$D(\mathbf{f}^{-1})(\mathbf{f}(\mathbf{x})) = D\mathbf{f}(\mathbf{x})^{-1} \qquad \text{for } \mathbf{x} \in V.$$

If $\mathbf{f}$ is smooth, so is $\mathbf{f}^{-1}$.

Sketch of proof. Replacing $\mathbf{f}(\mathbf{x})$ by $D\mathbf{f}(\mathbf{a})^{-1}(\mathbf{f}(\mathbf{a}+\mathbf{x}) - \mathbf{f}(\mathbf{a}))$, we may assume $\mathbf{a} = \mathbf{0}$, $\mathbf{f}(\mathbf{0}) = \mathbf{0}$ and $D\mathbf{f}(\mathbf{0}) = I$. Then $\mathbf{g}(\mathbf{x}) = \mathbf{x} - \mathbf{f}(\mathbf{x})$ has $D\mathbf{g}(\mathbf{0}) = 0$, so by continuity there is an $r > 0$ such that the closed ball $\overline{B}_r = \{\|\mathbf{x}\| \le r\}$ lies in U and $\|D\mathbf{g}(\mathbf{x})\|_2 \le \frac{1}{2}$ on it. By Lemma A.5.1, $\mathbf{g}$ is a contraction on $\overline{B}_r$ with $q = \frac{1}{2}$, and $\|\mathbf{g}(\mathbf{x})\| \le \frac{1}{2}\|\mathbf{x}\|$. For $\|\mathbf{y}\| < \frac{r}{2}$, the map $T(\mathbf{x}) = \mathbf{y} + \mathbf{g}(\mathbf{x})$ maps $\overline{B}_r$ into itself, since $\|T(\mathbf{x})\| \le \frac{r}{2} + \frac{r}{2}$, and it is a contraction. Its unique fixed point is the unique $\mathbf{x} \in \overline{B}_r$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$, and it satisfies $\|\mathbf{x}\| \le \|\mathbf{y}\| + \frac{1}{2}\|\mathbf{x}\|$, i.e., $\|\mathbf{x}\| \le 2\|\mathbf{y}\| < r$. So with $W = B(\mathbf{0}, \frac{r}{2})$ and $V = B(\mathbf{0}, r) \cap \mathbf{f}^{-1}(W)$, $\mathbf{f} : V \to W$ is bijective, and as in Corollary A.5.5, the inverse is continuous. One can show that it is also differentiable; then the chain rule applied to $\mathbf{f}^{-1}(\mathbf{f}(\mathbf{x})) = \mathbf{x}$ gives $D(\mathbf{f}^{-1})(\mathbf{f}(\mathbf{x}))\, D\mathbf{f}(\mathbf{x}) = I$. □

Example A.5.7.

1. Polar coordinates $\mathbf{f}(r, \varphi) = (r\cos\varphi, r\sin\varphi)$ have $\det D\mathbf{f} = r$ (Example 4.7 of Lecture 3). So near every point with $r > 0$, they have a smooth local inverse. But $\mathbf{f}$ is not injective, since $\mathbf{f}(r, \varphi + 2\pi) = \mathbf{f}(r, \varphi)$: the theorem is only local.
2. The map $F(x, y) = (x^2 - y^2, 2xy)$, i.e., $z \mapsto z^2$ for $z = x + iy$, has $\det DF = 4(x^2 + y^2)$. Near every point $z \neq 0$, it has a local inverse, a "branch of the square root". Globally, it is two-to-one, since $F(-z) = F(z)$, and at $z = 0$, it is not even locally injective.

A.6 The Implicit Function Theorem

When can an equation $F(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ be solved for some of the variables? We write points of $\mathbb{R}^{k+m}$ as $(\mathbf{x}, \mathbf{y})$ with $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^m$, and split the Jacobian of $F : \mathbb{R}^{k+m} \to \mathbb{R}^m$ into the $m \times k$ matrix $D_{\mathbf{x}}F$ and the square $m \times m$ matrix $D_{\mathbf{y}}F$ of partial derivatives with respect to $\mathbf{x}$ and $\mathbf{y}$.

Theorem A.6.1 (Implicit function theorem)**.** *Let $F : U \to \mathbb{R}^m$ be continuously differentiable on an open set $U \subseteq \mathbb{R}^{k+m}$, let $F(\mathbf{a}, \mathbf{b}) = \mathbf{0}$, and let $D_{\mathbf{y}}F(\mathbf{a}, \mathbf{b})$ be invertible. Then there are open sets $V \ni \mathbf{a}$ and $W \ni (\mathbf{a}, \mathbf{b})$ and a continuously differentiable function $\mathbf{g} : V \to \mathbb{R}^m$ with $\mathbf{g}(\mathbf{a}) = \mathbf{b}$, such that for $(\mathbf{x}, \mathbf{y}) \in W$,*

$$F(\mathbf{x}, \mathbf{y}) = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{x} \in V \text{ and } \mathbf{y} = \mathbf{g}(\mathbf{x}).$$

Moreover, $D\mathbf{g}(\mathbf{x}) = -D_{\mathbf{y}}F(\mathbf{x}, \mathbf{g}(\mathbf{x}))^{-1}\, D_{\mathbf{x}}F(\mathbf{x}, \mathbf{g}(\mathbf{x}))$.

Sketch of proof. Let $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, F(\mathbf{x}, \mathbf{y}))$. Its Jacobian $\begin{pmatrix} I & 0 \\ D_{\mathbf{x}}F & D_{\mathbf{y}}F \end{pmatrix}$ is invertible at $(\mathbf{a}, \mathbf{b})$: the equations $\mathbf{h} = \mathbf{u}$, $D_{\mathbf{x}}F\,\mathbf{h} + D_{\mathbf{y}}F\,\mathbf{k} = \mathbf{v}$ have the unique solution $\mathbf{h} = \mathbf{u}$, $\mathbf{k} = D_{\mathbf{y}}F^{-1}(\mathbf{v} - D_{\mathbf{x}}F\,\mathbf{u})$. By Theorem A.5.6, Φ has a local inverse near $\Phi(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{0})$. Since Φ does not change $\mathbf{x}$, neither does its inverse, which therefore has the form $(\mathbf{x}, \mathbf{z}) \mapsto (\mathbf{x}, G(\mathbf{x}, \mathbf{z}))$. Near $(\mathbf{a}, \mathbf{b})$, $F(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ means $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \mathbf{0})$, i.e., $\mathbf{y} = G(\mathbf{x}, \mathbf{0}) =: \mathbf{g}(\mathbf{x})$. Differentiating $F(\mathbf{x}, \mathbf{g}(\mathbf{x})) = \mathbf{0}$ with the chain rule gives $D_{\mathbf{x}}F + D_{\mathbf{y}}F\, D\mathbf{g} = 0$. □

Example A.6.2.

1. The unit circle $F(x, y) = x^2 + y^2 - 1 = 0$ has $\frac{\partial F}{\partial y} = 2y$. Near every point with $y \neq 0$, the circle is the graph of a function of x ($y = \pm\sqrt{1 - x^2}$), with $\frac{dy}{dx} = -\frac{2x}{2y}$. Near $(\pm 1, 0)$, it is not, but it is the graph of a function of y, since $\frac{\partial F}{\partial x} = 2x \neq 0$ there.

2. For the folium of Descartes $x^3+y^3=3xy$ (Example 9.4 of Lecture 3), $\frac{\partial F}{\partial y} = 3y^2 - 3x$ vanishes on the curve only at $(0,0)$, where the curve crosses itself, and at $(2^{2/3}, 2^{1/3})$, where its tangent is vertical. Near every other point of the curve, y is a smooth function of x.

Remark A.6.3 (Implicit differentiation in machine learning). Sometimes a quantity is defined as the solution of an equation, $G(\mathbf{z}, \boldsymbol{\theta}) = \mathbf{0}$, and we need its derivative with respect to $\boldsymbol{\theta}$. Examples are the minimizer $\mathbf{z}^*(\boldsymbol{\theta})$ of an inner optimization problem with hyperparameters $\boldsymbol{\theta}$ (here $G = \nabla_{\mathbf{z}}$ of the inner objective), and the fixed point $\mathbf{z} = \mathbf{h}(\mathbf{z}, \boldsymbol{\theta})$ of an "implicit layer" in deep equilibrium models. By Theorem A.6.1, $D\mathbf{z}^*(\boldsymbol{\theta}) = -D_{\mathbf{z}}G^{-1} D_{\boldsymbol{\theta}} G$, so we do not need to backpropagate through all the iterations of the solver that computed $\mathbf{z}^*$. For example, the ridge regression solution $\mathbf{w}^*(\lambda)$ solves $G(\mathbf{w}, \lambda) = X^\top(X\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w} = \mathbf{0}$, so

$$\frac{d\mathbf{w}^*}{d\lambda} = -(X^\top X + \lambda I)^{-1} \mathbf{w}^*(\lambda),$$

which can be used to tune λ by gradient descent on a validation loss.

A.7 Submanifolds and Tangent Spaces

A k-dimensional manifold is a set that looks like $\mathbb{R}^k$ near each of its points: a curve looks like a line, and a surface like a plane. We consider manifolds that lie in $\mathbb{R}^n$ and are described by equations.

Definition A.7.1 (Submanifold)**.** A set $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold of $\mathbb{R}^n$ if for every $\mathbf{p} \in M$, there are an open set $U \ni \mathbf{p}$ and a smooth map $F : U \to \mathbb{R}^{n-k}$ such that

$$M \cap U = \{\mathbf{x} \in U : F(\mathbf{x}) = \mathbf{0}\} \qquad \text{and} \qquad \operatorname{rank} DF(\mathbf{p}) = n - k.$$

(For $k = n$, there is no equation: M is an open subset of $\mathbb{R}^n$.)

So locally, M is cut out by $n-k$ independent equations. In particular, if $F : U \to \mathbb{R}^m$ is smooth and $\mathbf{c} \in \mathbb{R}^m$ is a regular value of F, i.e., $\operatorname{rank} DF(\mathbf{x}) = m$ for every $\mathbf{x}$ with $F(\mathbf{x}) = \mathbf{c}$, then the level set $F^{-1}(\mathbf{c})$ is an $(n-m)$-dimensional submanifold (use $F - \mathbf{c}$ in the definition). The implicit function theorem shows that submanifolds really look like $\mathbb{R}^k$.

Proposition A.7.2 (Submanifolds are local graphs)**.** *Let M be a k-dimensional submanifold and $\mathbf{p} \in M$. After renumbering the coordinates of $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k}$, there are open sets $V \subseteq \mathbb{R}^k$ and $W \ni \mathbf{p}$ and a smooth map $\mathbf{g} : V \to \mathbb{R}^{n-k}$ such that $M \cap W = \{(\mathbf{x}, \mathbf{g}(\mathbf{x})) : \mathbf{x} \in V\}$. In particular, $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{g}(\mathbf{x}))$ is a homeomorphism from the open set $V \subseteq \mathbb{R}^k$ onto $M \cap W$.*

Proof. The $(n-k) \times n$ matrix $DF(\mathbf{p})$ has rank $n-k$, so it has $n-k$ linearly independent columns (Lecture 1). Renumber the coordinates so that these are the last $n-k$ columns, and write $\mathbf{x} \in \mathbb{R}^n$ as $(\mathbf{x}', \mathbf{y})$. Then $D_{\mathbf{y}} F(\mathbf{p})$ is invertible, and Theorem A.6.1 gives $\mathbf{g}$. The inverse of $\mathbf{x}' \mapsto (\mathbf{x}', \mathbf{g}(\mathbf{x}'))$ is the projection to the first k coordinates, which is continuous. □

Example A.7.3.

1. The sphere $S^{n-1} = \{\mathbf{x} : \|\mathbf{x}\|^2 = 1\}$ is an $(n-1)$-dimensional submanifold, since $F(\mathbf{x}) = \|\mathbf{x}\|^2 - 1$ has $DF(\mathbf{x}) = 2\mathbf{x}^\top \neq \mathbf{0}$ on the sphere.

2. The graph $\{(\mathbf{x}, \mathbf{g}(\mathbf{x}))\}$ of a smooth $\mathbf{g} : \mathbb{R}^k \to \mathbb{R}^{n-k}$ is a k-dimensional submanifold: $F(\mathbf{x}, \mathbf{y}) = \mathbf{y} - \mathbf{g}(\mathbf{x})$ has $DF = (-D\mathbf{g},\, I)$, of rank $n-k$.

3. $\mathrm{GL}(n, \mathbb{R})$ is open in $\mathbb{R}^{n \times n}$, hence an n^2-dimensional submanifold.

4. The special linear group $\mathrm{SL}(n,\mathbb{R}) = \{X : \det X = 1\}$ is a submanifold of dimension $n^2 - 1$. By Jacobi's formula (Proposition 6.4 of Lecture 3), $D\det(X)H = \det(X)\operatorname{tr}(X^{-1}H)$, which for $\det X = 1$ and $H = X$ equals $n \neq 0$. So $D\det(X) \neq 0$, and 1 is a regular value of det.

5. The orthogonal group $\mathrm{O}(n) = \{Q : Q^\top Q = I\}$ is a submanifold of dimension $\frac{n(n-1)}{2}$. Let $F(Q) = Q^\top Q$, a map from $\mathbb{R}^{n\times n}$ to the space Sym_n of symmetric matrices, which has dimension $\frac{n(n+1)}{2}$. Its derivative is $DF(Q)H = H^\top Q + Q^\top H$. For $Q \in \mathrm{O}(n)$ and any symmetric S, the matrix $H = \frac{1}{2}QS$ satisfies $DF(Q)H = \frac{1}{2}SQ^\top Q + \frac{1}{2}Q^\top QS = S$, so $DF(Q)$ is surjective, i.e., has full rank $\frac{n(n+1)}{2}$, and $\mathrm{O}(n)$ has dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$. For $n = 2$, this is 1 (one rotation angle), and for $n = 3$, it is 3 (three angles, as for the rotations R_1, R_2, R_3 of Lecture 2).

Not every level set is a submanifold. The level set $\{x^2 - y^2 = 0\}$ of Figure 1 in Lecture 3 consists of two lines that cross at the origin, where the gradient $(2x, -2y)$ vanishes. It is not a 1-dimensional submanifold near the origin: removing the origin from a small piece of it leaves four connected pieces, while removing a point from a small piece of a curve that looks like a line leaves two. Similarly, the double cone $\{x^2 + y^2 = z^2\}$ is not a submanifold at its tip. On the other hand, the rank condition is sufficient, but not necessary: $\{x^2 = 0\}$ is the y-axis, a perfectly good submanifold, although $D(x^2) = (2x, 0)$ vanishes on it.

Definition A.7.4 (Tangent space)**.** Let M be a submanifold and $\mathbf{p} \in M$. The tangent space $T_\mathbf{p}M$ is the set of velocities $\boldsymbol{\gamma}'(0)$ of all smooth curves $\boldsymbol{\gamma} : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$ with $\boldsymbol{\gamma}(t) \in M$ for all t and $\boldsymbol{\gamma}(0) = \mathbf{p}$. Its orthogonal complement $N_\mathbf{p}M = (T_\mathbf{p}M)^\perp$ is the normal space.

Theorem A.7.5. *Let M be a k-dimensional submanifold, and let $M \cap U = F^{-1}(\mathbf{0})$ near $\mathbf{p}$ as in Definition A.7.1. Then*

$$T_\mathbf{p}M = \operatorname{null} DF(\mathbf{p}) \qquad \text{and} \qquad N_\mathbf{p}M = \operatorname{row} DF(\mathbf{p}) = \operatorname{span}\{\nabla F_1(\mathbf{p}), \ldots, \nabla F_{n-k}(\mathbf{p})\}.$$

In particular, $T_\mathbf{p}M$ is a k-dimensional subspace.

Proof. If $\boldsymbol{\gamma}$ is a curve in M through $\mathbf{p}$, then $F(\boldsymbol{\gamma}(t)) = \mathbf{0}$ for small t, and the chain rule gives $DF(\mathbf{p})\boldsymbol{\gamma}'(0) = \mathbf{0}$. So $T_\mathbf{p}M \subseteq \operatorname{null} DF(\mathbf{p})$, which has dimension $n - (n-k) = k$ by the rank–nullity theorem (Lecture 1). Conversely, write M near $\mathbf{p} = (\mathbf{a}, \mathbf{g}(\mathbf{a}))$ as a graph as in Proposition A.7.2. For every $\mathbf{v} \in \mathbb{R}^k$, the curve $t \mapsto (\mathbf{a} + t\mathbf{v}, \mathbf{g}(\mathbf{a} + t\mathbf{v}))$ lies in M and has velocity $(\mathbf{v}, D\mathbf{g}(\mathbf{a})\mathbf{v})$ at $t = 0$. These velocities form a k-dimensional subspace (the image of an injective linear map), so $T_\mathbf{p}M$ contains a k-dimensional subspace of the k-dimensional space $\operatorname{null} DF(\mathbf{p})$, and the two are equal. Finally, $\operatorname{null}(A)^\perp = \operatorname{row}(A)$ for every matrix A (Lecture 2), and the rows of $DF(\mathbf{p})$ are the transposed gradients $\nabla F_i(\mathbf{p})^\top$. □

Example A.7.6. 1. $T_\mathbf{p}S^{n-1} = \operatorname{null}(2\mathbf{p}^\top) = \{\mathbf{v} : \mathbf{p}^\top\mathbf{v} = 0\} = \mathbf{p}^\perp$: the tangent vectors of a sphere are orthogonal to the radius (Figure 1).

2. $T_I\mathrm{SL}(n,\mathbb{R}) = \{H : D\det(I)H = 0\} = \{H : \operatorname{tr} H = 0\}$, the traceless matrices: to first order, $\det(I + tH) = 1 + t\operatorname{tr} H$ (Lemma 6.3 of Lecture 3), so a matrix curve through I keeps determinant 1 only if its velocity has trace 0.

3. $T_Q\mathrm{O}(n) = \{H : H^\top Q + Q^\top H = 0\}$. With $K = Q^\top H$, the condition says $K^\top + K = 0$, so $T_Q\mathrm{O}(n) = \{QK : K^\top = -K\}$, and $T_I\mathrm{O}(n)$ is the space of skew-symmetric matrices, of dimension $\frac{n(n-1)}{2}$. For example, the rotations $R(t)$ by the angle t in the plane have $R'(0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

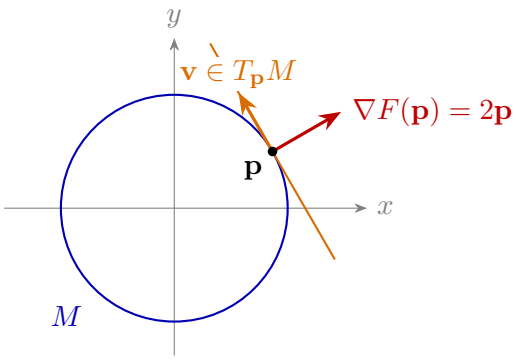


Figure 1: The circle $M = \{F = 0\}$ with $F(x, y) = x^2 + y^2 - 1$. The tangent space $T_{\mathbf{p}}M = \operatorname{null} DF(\mathbf{p}) = \mathbf{p}^{\perp}$ is the tangent line (shifted to $\mathbf{p}$), and the normal space is spanned by the gradient $\nabla F(\mathbf{p})$.

A.8 Lagrange Multipliers and Optimization on Manifolds

We can now complete the proof of the Lagrange multiplier theorem (Theorem 12.1 of Lecture 3).

Theorem A.8.1 (Lagrange multipliers)**.** *Let f and $g_1, \ldots, g_m$ be smooth on an open set $U \subseteq \mathbb{R}^n$, let $\mathbf{x}^*$ be a local minimum of f on $M = \{\mathbf{x} \in U : g_1(\mathbf{x}) = \cdots = g_m(\mathbf{x}) = 0\}$, and let $\nabla g_1(\mathbf{x}^*), \ldots, \nabla g_m(\mathbf{x}^*)$ be linearly independent. Then $\nabla f(\mathbf{x}^*) \in \operatorname{span}\{\nabla g_1(\mathbf{x}^*), \ldots, \nabla g_m(\mathbf{x}^*)\}$, i.e., there are λ_i with $\nabla f(\mathbf{x}^*) + \sum_i \lambda_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}$.*

Proof. The linear independence says that $DG(\mathbf{x}^*)$ has rank m for $G = (g_1, \ldots, g_m)$, so near $\mathbf{x}^*$, M is an $(n - m)$-dimensional submanifold (the rank stays m near $\mathbf{x}^*$, since a nonzero $m \times m$ minor stays nonzero by continuity). For every $\mathbf{v} \in T_{\mathbf{x}^*}M$, choose a curve $\boldsymbol{\gamma}$ in M with $\boldsymbol{\gamma}(0) = \mathbf{x}^*$ and $\boldsymbol{\gamma}'(0) = \mathbf{v}$. The function $t \mapsto f(\boldsymbol{\gamma}(t))$ has a local minimum at $t = 0$, so its derivative $\nabla f(\mathbf{x}^*)^{\top}\mathbf{v}$ is 0. Hence $\nabla f(\mathbf{x}^*) \in (T_{\mathbf{x}^*}M)^{\perp} = N_{\mathbf{x}^*}M = \operatorname{span}\{\nabla g_i(\mathbf{x}^*)\}$ by Theorem A.7.5. □

The proof suggests how to do calculus on a manifold: only the component of ∇f in the tangent space matters.

Definition A.8.2 (Riemannian gradient)**.** Let M be a submanifold of $\mathbb{R}^n$, $\mathbf{p} \in M$, and f smooth near M. The Riemannian gradient of f on M at $\mathbf{p}$ is the orthogonal projection (Lecture 2) of $\nabla f(\mathbf{p})$ onto $T_{\mathbf{p}}M$:

$$\operatorname{grad}_M f(\mathbf{p}) = P_{\mathbf{p}} \nabla f(\mathbf{p}).$$

Since $\nabla f(\mathbf{p}) - \operatorname{grad}_M f(\mathbf{p})$ is orthogonal to $T_{\mathbf{p}}M$, we have $\nabla f(\mathbf{p})^{\top}\mathbf{v} = \operatorname{grad}_M f(\mathbf{p})^{\top}\mathbf{v}$ for all tangent vectors $\mathbf{v}$: the Riemannian gradient gives the rates of change of f along M. As in Theorem 3.9 of Lecture 3, it points in the direction of steepest ascent among the tangent directions, and by Theorem A.8.1, $\operatorname{grad}_M f(\mathbf{p}) = \mathbf{0}$ exactly at the points that satisfy the Lagrange condition. These points are the critical points of f on M.

Riemannian gradient descent minimizes f on M by taking a step in the direction $-\operatorname{grad}_M f(\mathbf{p}_k)$, which leaves M slightly, and mapping the result back to M with a retraction R:

$$\mathbf{p}_{k+1} = R(\mathbf{p}_k - \eta \operatorname{grad}_M f(\mathbf{p}_k)).$$

On the sphere, $P_{\mathbf{p}} = I - \mathbf{p}\mathbf{p}^{\top}$, so $\operatorname{grad}_M f(\mathbf{p}) = \nabla f(\mathbf{p}) - (\mathbf{p}^{\top}\nabla f(\mathbf{p}))\mathbf{p}$, and the retraction can simply normalize: $R(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$. On $\mathrm{O}(n)$, one can retract with the Q factor of the QR decomposition

(Lecture 2). Such methods are used for PCA, for orthogonality constraints on weight matrices, and for low-rank matrix problems.

Example A.8.3 (The Rayleigh quotient on the sphere)**.** Let A be symmetric with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and unit eigenvectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$, and let $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on S^{n-1}. Then $\operatorname{grad}_M f(\mathbf{x}) = 2A\mathbf{x} - 2(\mathbf{x}^\top A\mathbf{x})\mathbf{x}$, which is $\mathbf{0}$ if and only if $\mathbf{x}$ is an eigenvector: the critical points are the $2n$ points $\pm\mathbf{q}_i$. To see their types, move from $\mathbf{q}_i$ towards $\mathbf{q}_j$ along the great circle $\boldsymbol{\gamma}(t) = \cos t\,\mathbf{q}_i + \sin t\,\mathbf{q}_j$:

$$f(\boldsymbol{\gamma}(t)) = \lambda_i \cos^2 t + \lambda_j \sin^2 t = \lambda_i + (\lambda_j - \lambda_i)\sin^2 t.$$

So f decreases in the $n - i$ tangent directions $\mathbf{q}_j$ with $\lambda_j < \lambda_i$, and increases in the $i - 1$ directions with $\lambda_j > \lambda_i$. Hence $\pm\mathbf{q}_1$ are the maxima, $\pm\mathbf{q}_n$ the minima, and all other eigenvectors are saddle points. In particular, Riemannian gradient ascent for the first principal component (Lecture 2) has exactly one pair of local maxima, the correct answer; the other critical points are saddle points, which gradient ascent avoids from almost every starting point.

A.9 Critical Points and Morse Theory

A.9.1 Regular and critical values

Definition A.9.1 (Critical points and values of maps)**.** Let $F : U \to \mathbb{R}^m$ be smooth on an open set $U \subseteq \mathbb{R}^n$. A point $\mathbf{x}$ is a critical point of F if $\operatorname{rank} DF(\mathbf{x}) < m$, and a value $\mathbf{c}$ is a critical value if $\mathbf{c} = F(\mathbf{x})$ for a critical point $\mathbf{x}$. All other values in $\mathbb{R}^m$ are regular values.

For $m = 1$, the critical points are the points with $\nabla F(\mathbf{x}) = \mathbf{0}$, as in Lecture 3. By Section A.7, the level set of a regular value is a submanifold (or empty). How many critical values can there be? A set $Z \subseteq \mathbb{R}^m$ has measure zero if for every $\varepsilon > 0$ it can be covered by countably many boxes whose volumes add up to less than ε. For example, finite and countable sets, and lines in $\mathbb{R}^2$, have measure zero.

Theorem A.9.2 (Sard)**.** *The set of critical values of a smooth map $F : U \to \mathbb{R}^m$ has measure zero.*

We do not prove this theorem. It says that almost every value is regular, so almost every level set of a smooth function $f : U \to \mathbb{R}$ is an $(n-1)$-dimensional submanifold or empty. Note that the theorem is about values, not points: a constant function has only critical points, but only one critical value.

A.9.2 Nondegenerate critical points

From now on, let $f : U \to \mathbb{R}$ be smooth. Lecture 3 classified critical points by the eigenvalues of the Hessian (Theorem 9.3). The test was inconclusive when an eigenvalue is 0.

Definition A.9.3 (Nondegenerate critical points and the index)**.** A critical point $\mathbf{p}$ of f is nondegenerate if the Hessian $\nabla^2 f(\mathbf{p})$ is invertible, i.e., has no eigenvalue 0. Its index is the number of negative eigenvalues of $\nabla^2 f(\mathbf{p})$, counted with multiplicity. A function whose critical points are all nondegenerate is a Morse function.

So a nondegenerate critical point of index 0 is a strict local minimum, one of index n is a strict local maximum, and all others are saddle points. The index counts the independent directions in which f decreases. For example, $x^3 + y^3 - 3xy$ (Example 9.4 of Lecture 3) is a Morse function,

with a critical point of index 1 at the origin and one of index 0 at $(1, 1)$. The "monkey saddle" $x^3 - 3xy^2$ has a degenerate critical point at the origin, where its Hessian is 0; near the origin, it goes up and down three times, unlike any nondegenerate saddle.

Proposition A.9.4. *Nondegenerate critical points are isolated: each one has a neighborhood that contains no other critical point.*

Proof. Let $H = \nabla^2 f(\mathbf{p})$, and let $\mu > 0$ be the smallest absolute value of an eigenvalue of H. Writing $\mathbf{h}$ in an orthonormal eigenbasis (Lecture 2) shows $\|H\mathbf{h}\| \geq \mu\|\mathbf{h}\|$. The gradient is differentiable with derivative H at $\mathbf{p}$, so $\nabla f(\mathbf{p} + \mathbf{h}) = H\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\| \leq \frac{\mu}{2}\|\mathbf{h}\|$ for small $\mathbf{h}$. Then $\|\nabla f(\mathbf{p} + \mathbf{h})\| \geq \frac{\mu}{2}\|\mathbf{h}\| > 0$ for small $\mathbf{h} \neq \mathbf{0}$. □

In Lecture 3, we saw that near a critical point, f is approximately a sum of parabolas, $f(\mathbf{p}) + \frac{1}{2}\sum_i \lambda_i y_i^2$, in the coordinates given by the eigenvectors of the Hessian. For nondegenerate critical points, this becomes exact after a nonlinear change of coordinates.

Theorem A.9.5 (Morse lemma)**.** *Let $\mathbf{p}$ be a nondegenerate critical point of f with index k. Then there is a diffeomorphism $\boldsymbol{\varphi}$ from a neighborhood of $\mathbf{p}$ onto a neighborhood of $\mathbf{0}$, with $\boldsymbol{\varphi}(\mathbf{p}) = \mathbf{0}$, such that in the coordinates $\mathbf{y} = \boldsymbol{\varphi}(\mathbf{x})$,*

$$f = f(\mathbf{p}) - y_1^2 - \cdots - y_k^2 + y_{k+1}^2 + \cdots + y_n^2.$$

We do not prove the Morse lemma. It says that near a nondegenerate critical point, the higher-order terms of the Taylor expansion do not change the picture at all: up to a smooth deformation, the function is a bowl, a saddle with k downward directions, or an upside-down bowl. Degenerate critical points can be much more complicated, but they are rare.

Theorem A.9.6 (Morse functions are generic)**.** *Let $f : U \to \mathbb{R}$ be smooth. For almost every $\mathbf{a} \in \mathbb{R}^n$ (all except a set of measure zero), the function $f_{\mathbf{a}}(\mathbf{x}) = f(\mathbf{x}) - \mathbf{a}^\top \mathbf{x}$ is a Morse function.*

Proof. We have $\nabla f_{\mathbf{a}} = \nabla f - \mathbf{a}$ and $\nabla^2 f_{\mathbf{a}} = \nabla^2 f$. So $\mathbf{x}$ is a critical point of $f_{\mathbf{a}}$ if and only if $\nabla f(\mathbf{x}) = \mathbf{a}$, and it is degenerate if and only if $\nabla^2 f(\mathbf{x})$, the Jacobian of the map $\nabla f : U \to \mathbb{R}^n$, is singular. Hence $f_{\mathbf{a}}$ has a degenerate critical point if and only if $\mathbf{a}$ is a critical value of ∇f, and by Sard's theorem, these $\mathbf{a}$ form a set of measure zero. □

For example, $f(x) = x^3$ has a degenerate critical point at 0, but $x^3 - ax$ is a Morse function for every $a \neq 0$: it has two nondegenerate critical points if $a > 0$, and none if $a < 0$. An arbitrarily small, random linear perturbation removes all degenerate critical points.

A.9.3 Critical points and the shape of sublevel sets

Morse theory studies how the sublevel sets $M^c = \{\mathbf{x} : f(\mathbf{x}) \leq c\}$ change as c increases. We state its two main results informally, for a Morse function f whose sublevel sets are compact (for example, a Morse function on a compact manifold, or a coercive Morse function on $\mathbb{R}^n$).

- If there are no critical values in $[a, b]$, then M^a and M^b have the same shape: M^b can be continuously deformed into M^a, by flowing downhill along $-\nabla f$. Nothing interesting happens between critical values.
- When c passes the value of a critical point of index k, the sublevel set changes by attaching a k-dimensional "cell" (a point for $k = 0$, a segment for $k = 1$, a disk for $k = 2$, and so on). A minimum ($k = 0$) creates a new connected component. A saddle point of index 1 attaches a bridge, which either connects two components or closes a loop.

Example A.9.7 (A double well)**.** The function $f(x, y) = (x^2 - 1)^2 + y^2$ has $\nabla f = (4x(x^2 - 1), 2y)$, so its critical points are $(\pm 1, 0)$, with $f = 0$ and $\nabla^2 f = \operatorname{diag}(8, 2)$ (index 0), and $(0, 0)$, with $f = 1$ and $\nabla^2 f = \operatorname{diag}(-4, 2)$ (index 1). For $c < 0$, the sublevel set is empty; at $c = 0$, two components appear at the minima; at $c = 1$, the saddle point attaches a bridge that joins them, and for $c > 1$, the sublevel set is connected (Figure 2). For gradient descent, this means that the two basins of the minima are separated by the saddle: an iteration that starts with $f(\mathbf{x}_0) < 1$ and decreases f in every step cannot move from one basin to the other.

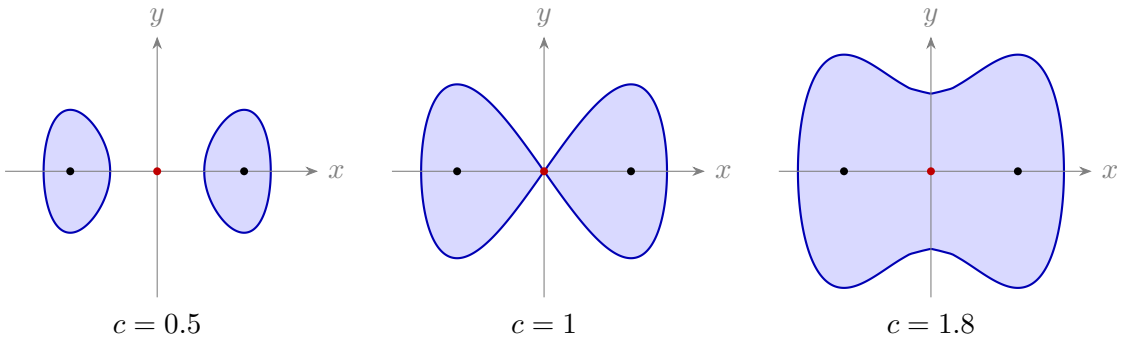


Figure 2: Sublevel sets $\{f \leq c\}$ of the double well $f(x, y) = (x^2 - 1)^2 + y^2$, with the two minima (black, index 0, value 0) and the saddle point (red, index 1, value 1). Below the critical value 1, the sublevel set has two components; at $c = 1$, they touch at the saddle point; above it, they have merged into one.

On a compact manifold, these changes must add up. The Euler characteristic $\chi(M)$ is a number that only depends on the topology of M: for a surface divided into V vertices, E edges and F faces, it is $V - E + F$, which does not depend on the subdivision. For example, the sphere has $\chi = 2$ (a cube has $8 - 12 + 6 = 2$), and the torus (the surface of a doughnut) has $\chi = 0$.

Theorem A.9.8 (Morse relation)**.** *Let f be a Morse function on a compact manifold M, and let c_k be the number of its critical points of index k. Then*

$$\chi(M) = c_0 - c_1 + c_2 - \cdots + (-1)^n c_n.$$

We do not prove this theorem; it follows from the fact that attaching a k-cell changes the Euler characteristic by $(-1)^k$. It shows that the topology of M forces critical points to exist, whatever function we choose.

Example A.9.9.

1. The height function on the sphere S^2 has a minimum and a maximum: $\chi(S^2) = 1 - 0 + 1 = 2$.
2. The height function on an upright torus has a minimum, two saddle points and a maximum (Figure 3): $\chi = 1 - 2 + 1 = 0$. In fact, every Morse function on the torus has at least four critical points, at least two of which are saddle points (this follows from the Morse inequalities, which we do not discuss).
3. For the Rayleigh quotient on S^{n-1} (Example A.8.3), the two critical points $\pm\mathbf{q}_i$ have index $n - i$, so $\sum_k (-1)^k c_k = 2\sum_{i=1}^{n} (-1)^{n-i}$, which is 2 for odd n and 0 for even n. This agrees with the Euler characteristic of the sphere S^{n-1}, which is 2 if $n - 1$ is even and 0 if $n - 1$ is odd. For $n = 3$ and $A = \operatorname{diag}(3, 2, 1)$: two maxima $\pm\mathbf{e}_1$, two saddle points $\pm\mathbf{e}_2$, two minima $\pm\mathbf{e}_3$, and $2 - 2 + 2 = 2$.

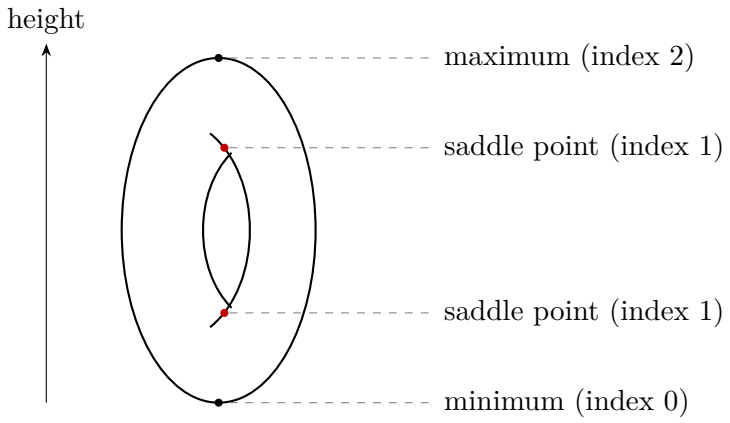


Figure 3: The height function on an upright torus has four critical points: the bottom (a minimum), the bottom and the top of the hole (two saddle points), and the top (a maximum). As the height c increases, the sublevel set is first a disk around the bottom (after the minimum), then a bent tube, i.e., a cylinder (after the first saddle point, where the disk closes up around the tube), then a torus with a hole at its top (after the second saddle point, where the two ends of the tube meet and close a loop around the hole of the torus), and finally the whole torus (after the maximum).

A.10 Topology in Machine Learning

We end with a few places where the ideas of this appendix appear in machine learning. Some of them are theorems, others are well-supported heuristics and active research topics.

- **The manifold hypothesis.** High-dimensional data, such as images with millions of pixels, are believed to lie close to submanifolds of much smaller dimension: the images of a face, for example, vary with a small number of parameters such as pose, lighting and expression. PCA finds the best approximating linear subspace (Lecture 2); autoencoders and other nonlinear methods try to find curved, manifold-like structures.

- **Normalizing flows and topology.** A normalizing flow maps a simple distribution, such as a Gaussian on $\mathbb{R}^n$, to a complicated one by a diffeomorphism (Corollary A.5.5 and Remark 4.8 of Lecture 3). Diffeomorphisms preserve topology. So, by Proposition A.4.2, a flow cannot map $\mathbb{R}^n$, which is connected, exactly onto a distribution supported on two separate pieces, such as two clusters: it can only approximate it by squeezing a little probability mass through thin bridges between them.

- **Symmetries make critical points degenerate.** If a loss is invariant under a continuous family of transformations of the parameters, its minima are not isolated. For example, the ReLU satisfies $\max(0, ct) = c\max(0, t)$ for $c > 0$, so multiplying the incoming weights of a hidden unit by c and its outgoing weights by $\frac{1}{c}$ does not change the network. The set of minima then contains curves (in the simplest case, the hyperbola $w_1 w_2 = 1$ for the loss $(w_1 w_2 - 1)^2$), and by Proposition A.9.4, the Hessian at a minimum must have the eigenvalue 0, in the tangent directions of these curves. Loss functions of neural networks are therefore not Morse functions.

- **Saddle points in high dimensions.** At a critical point of a loss with millions of parameters, all eigenvalues of the Hessian must be nonnegative for a local minimum. A heuristic, supported

by experiments and by random matrix theory, says that critical points with high loss usually have many negative eigenvalues, i.e., a large index: they are saddle points rather than bad local minima. Gradient descent with noise, as in SGD, tends to escape from saddle points, which is part of the explanation why training works.

- **Connected sublevel sets.** Experiments show that two minima found by training the same network twice can often be joined by a path along which the loss stays low ("mode connectivity"): the sublevel sets $\{L \le c\}$ for small c seem to be connected, up to the symmetries of the network. Morse theory gives a language for such statements: a sublevel set can only become connected by passing saddle points of index 1.

A.11 Banach Spaces

In Lectures 2 and 4, we met vector spaces whose elements are functions, such as polynomials, continuous functions and random variables. These spaces are infinite-dimensional, and Remark A.3.9 showed that some results of the previous sections fail in them. Functional analysis studies such spaces. Its starting point is the property that makes $\mathbb{R}^n$ so convenient for analysis: completeness (Corollary A.3.2).

Definition A.11.1 (Normed and Banach spaces)**.** A norm on a real vector space V is a function $\|\cdot\| : V \to [0,\infty)$ such that $\|v\| = 0$ only for $v = 0$, $\|cv\| = |c|\|v\|$ and $\|u+v\| \le \|u\| + \|v\|$ for all $u, v \in V$ and $c \in \mathbb{R}$. A normed space is a metric space with the metric $d(u,v) = \|u - v\|$. It is a Banach space if it is complete, i.e., if every Cauchy sequence in it converges.

By Corollary A.3.2 and Theorem A.3.8, every finite-dimensional normed space is a Banach space, since a basis identifies it with $\mathbb{R}^n$.

Example A.11.2 (Banach spaces)**.**

1. The space $C[a,b]$ of continuous functions $f : [a,b] \to \mathbb{R}$ with the maximum norm $\|f\|_\infty = \max_{a \le x \le b} |f(x)|$ is a Banach space (Proposition A.11.3 below). Convergence in this norm is uniform convergence: $\|f_k - f\|_\infty \to 0$ means that $f_k(x) \to f(x)$ at the same speed for all x.

2. For $1 \le p < \infty$, the sequences $x = (x_1, x_2, \ldots)$ with $\|x\|_p = (\sum_k |x_k|^p)^{1/p} < \infty$ form a Banach space ℓ^p, and the bounded sequences with $\|x\|_\infty = \sup_k |x_k|$ form a Banach space ℓ^∞.

3. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $1 \le p < \infty$. The measurable functions with $\|f\|_p = (\int |f|^p \, d\mu)^{1/p} < \infty$, where functions that agree almost everywhere are identified, form a Banach space $L^p(\mu)$ (the Riesz–Fischer theorem, which we do not prove). The integral is the Lebesgue integral of Lecture 4 (Definition 3.1). For a probability measure, these are the spaces L^p of random variables of Lecture 4; for the Lebesgue measure on $[a,b]$, we write $L^p[a,b]$; and for the counting measure on $\{1,2,\ldots\}$, $L^p(\mu)$ is ℓ^p.

4. Not every normed space is complete. On $C[0,1]$, $\|f\|_1 = \int_0^1 |f(x)|\,dx$ is also a norm. The ramps f_k that are 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2} + \frac{1}{k}, 1]$ and linear in between satisfy $\|f_k - f_l\|_1 \le \frac{1}{2\min(k,l)}$, so they form a Cauchy sequence for this norm. But they converge in this norm to the step function $\mathbf{1}_{(1/2,1]}$, and no continuous function is their limit. So $(C[0,1], \|\cdot\|_1)$ is not complete. Adding the limits of all Cauchy sequences, as the real numbers complete the rational numbers, gives $L^1[0,1]$.

Proposition A.11.3. *$C[a,b]$ with the maximum norm is complete.*

Proof. Let (f_k) be a Cauchy sequence. For each x, $|f_k(x) - f_l(x)| \le \|f_k - f_l\|_\infty$, so $(f_k(x))$ is a Cauchy sequence of real numbers, with a limit $f(x)$. Given $\varepsilon > 0$, choose N with $\|f_k - f_l\|_\infty \le \varepsilon$ for $k, l \ge N$. Letting $l \to \infty$ gives $|f_k(x) - f(x)| \le \varepsilon$ for all x and $k \ge N$, so $f_k \to f$ uniformly. It remains to show that f is continuous. Fix x, and choose $\delta > 0$ with $|f_N(y) - f_N(x)| < \varepsilon$ for $|y - x| < \delta$. Then for $|y - x| < \delta$,

$$|f(y) - f(x)| \le |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)| < 3\varepsilon. \qquad \square$$

In particular, a uniform limit of continuous functions is continuous. A pointwise limit need not be: on $[0, 1]$, x^k converges to the function that is 0 for $x < 1$ and 1 for $x = 1$.

Fixed points and differential equations. The proof of the Banach fixed-point theorem (Theorem A.5.3) only used the triangle inequality and completeness. So the theorem holds, with the same proof, for every contraction $T : C \to C$ of a closed nonempty subset C of a Banach space. This proves the existence theorem for differential equations of Lecture 3.

Theorem A.11.4 (Picard–Lindelöf)**.** *Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz continuous, $\|F(\mathbf{x}) - F(\mathbf{y})\| \le L\|\mathbf{x} - \mathbf{y}\|$, and let $T > 0$. Then for every $\mathbf{x}_0$, the ODE $\mathbf{x}' = F(\mathbf{x})$ has exactly one solution on $[0, T]$ with $\mathbf{x}(0) = \mathbf{x}_0$. Consequently, it has exactly one solution defined for all $t \in \mathbb{R}$ (Theorem 15.2 of Lecture 3).*

Proof. By the fundamental theorem of calculus, a continuous function $\mathbf{x} : [0, T] \to \mathbb{R}^n$ is a solution if and only if it is a fixed point of the map

$$(\Phi\mathbf{x})(t) = \mathbf{x}_0 + \int_0^t F(\mathbf{x}(s))\, ds,$$

which maps the space $C([0, T], \mathbb{R}^n)$ of continuous functions into itself. We give this space the weighted norm $\|\mathbf{x}\|_w = \max_{0 \le t \le T} e^{-2Lt}\|\mathbf{x}(t)\|$. Since $e^{-2LT} \le e^{-2Lt} \le 1$, it is equivalent to the maximum norm, and the space is complete, as in Proposition A.11.3. For continuous $\mathbf{x}$ and $\mathbf{y}$, and $0 \le t \le T$,

$$\|(\Phi\mathbf{x})(t) - (\Phi\mathbf{y})(t)\| \le \int_0^t L\|\mathbf{x}(s) - \mathbf{y}(s)\|\, ds \le \|\mathbf{x} - \mathbf{y}\|_w \int_0^t Le^{2Ls}\, ds \le \frac{e^{2Lt}}{2}\|\mathbf{x} - \mathbf{y}\|_w.$$

Multiplying by e^{-2Lt} gives $\|\Phi\mathbf{x} - \Phi\mathbf{y}\|_w \le \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|_w$. So Φ is a contraction, and it has exactly one fixed point. By uniqueness, the solutions on intervals $[0, T]$ of different lengths agree where they are both defined, so they form one solution on $[0, \infty)$. For negative times, apply the same argument to $\mathbf{y}(t) = \mathbf{x}(-t)$, which solves $\mathbf{y}' = -F(\mathbf{y})$. $\square$

With the maximum norm, the same estimate only shows that Φ is a contraction if $LT < 1$; the weight e^{-2Lt} removes this restriction. The iteration $\mathbf{x}_{k+1} = \Phi\mathbf{x}_k$ is called Picard iteration. For $x' = x$ with $x(0) = 1$, starting from the constant $x_0(t) = 1$, it gives $x_1(t) = 1 + t$, $x_2(t) = 1 + t + \frac{t^2}{2}$, and in general the Taylor polynomials of the solution e^t.

Bounded operators. Linear maps between infinite-dimensional spaces need not be continuous. A linear map $A : V \to W$ between normed spaces is bounded if its operator norm

$$\|A\| = \sup_{\|v\| \le 1} \|Av\|$$

is finite, and then $\|Av\| \le \|A\|\|v\|$ for all v, as for the matrix norm of Lecture 2. A linear map is continuous if and only if it is bounded: if it is bounded, then $\|Au - Av\| \le \|A\|\|u - v\|$; if it is continuous at 0, there is a $\delta > 0$ with $\|Av\| \le 1$ for $\|v\| \le \delta$, so $\|A\| \le \frac{1}{\delta}$. In finite dimensions, every linear map is bounded.

Example A.11.5. 1. (Integration.) The Volterra operator $(Vf)(x) = \int_0^x f(t)\,dt$ on $C[0,1]$ with the maximum norm satisfies $|(Vf)(x)| \le x\|f\|_\infty$, so $\|V\| = 1$, with equality for $f = 1$. By induction, $|(V^k f)(x)| \le \frac{x^k}{k!}\|f\|_\infty$, so $\|V^k\| = \frac{1}{k!}$: the powers of V decay much faster than $\|V\|^k = 1$.

2. (Differentiation.) On the polynomials on $[0,1]$ with the maximum norm, the derivative D is linear but not bounded: the polynomials x^k have the norm 1, but their derivatives kx^{k-1} have the norm k. Differentiation amplifies small wiggles of high frequency, which is why numerical differentiation by finite differences (Proposition 2.9 of Lecture 3) is sensitive to noise.

Dual spaces. The bounded linear functionals $\varphi : V \to \mathbb{R}$ on a normed space V form its dual space V^*, a Banach space with the operator norm. On $\mathbb{R}^n$, every linear functional is $\mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$. The Hahn–Banach theorem, which we do not prove, guarantees that every normed space has many bounded linear functionals: for every closed subspace $U \neq V$, there is a $\varphi \in V^*$ with $\varphi \neq 0$ and $\varphi(u) = 0$ for all $u \in U$. The dual space of $C[a,b]$ consists of the integrals $f \mapsto \int f\,d\mu$ with respect to signed measures μ, i.e., differences of two finite measures on the Borel sets (the Riesz–Markov theorem). For example, the evaluation $f \mapsto f(x_0) = \int f\,d\delta_{x_0}$ at a point is a bounded linear functional on $C[a,b]$ with the norm 1. On $L^2[a,b]$, evaluation at a point is not even defined, since the elements of L^2 are only defined up to null sets. The dual of a Hilbert space is the space itself, as we show in the next section.

A.12 Hilbert Spaces

Definition A.12.1 (Hilbert space)**.** An inner product space is a real vector space H with an inner product $\langle \cdot, \cdot \rangle$ (Lecture 2) and the norm $\|v\| = \sqrt{\langle v, v \rangle}$. It is a Hilbert space if it is complete with this norm.

The Cauchy–Schwarz inequality $|\langle u, v \rangle| \le \|u\|\|v\|$ and Pythagoras' theorem hold as in Lecture 2, since their proofs only use the properties of an inner product, and so does the parallelogram law $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$ (expand both squares). By the Cauchy–Schwarz inequality, the inner product is continuous.

Example A.12.2 (Hilbert spaces)**.** 1. $\mathbb{R}^n$ with the dot product, and ℓ^2 with $\langle x, y \rangle = \sum_k x_k y_k$, which converges by the Cauchy–Schwarz inequality.

2. $L^2(\mu)$ with $\langle f, g \rangle = \int fg\,d\mu$, for example $L^2[a,b]$ with $\langle f, g \rangle = \int_a^b f(x)g(x)\,dx$. For a probability space, it is the space L^2 of random variables with finite variance and the inner product $\langle X, Y \rangle = \mathbb{E}[XY]$ of Lecture 4.

3. None of the other spaces ℓ^p is a Hilbert space, since their norms do not satisfy the parallelogram law; in ℓ^1, for example, the unit sequences $e_1 = (1, 0, 0, \ldots)$ and $e_2 = (0, 1, 0, \ldots)$ give $4 + 4 \neq 2 + 2$. And $C[0,1]$ with the inner product of $L^2[0,1]$ is an inner product space that is not complete, by the argument of Example A.11.2(4).

The most important property of Hilbert spaces is that we can project onto closed subspaces, as in Lecture 2.

Theorem A.12.3 (Projection theorem)**.** *Let C be a nonempty closed convex subset of a Hilbert space H. Then every $x \in H$ has exactly one closest point $P_C x$ in C. If $C = U$ is a closed subspace, then $P_U x$ is the only $p \in U$ with $x - p \perp U$, and every $x \in H$ can be written in exactly one way as $x = u + w$ with $u \in U$ and $w \perp U$.*

Proof. Let $d = \inf_{c \in C} \|x - c\|$, and choose $c_k \in C$ with $\|x - c_k\| \to d$. The parallelogram law for $x - c_k$ and $x - c_l$ gives

$$\|c_k - c_l\|^2 = 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4\Big\|x - \frac{c_k + c_l}{2}\Big\|^2 \le 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4d^2,$$

since $\frac{c_k + c_l}{2} \in C$ by convexity. The right side tends to 0 as $k, l \to \infty$, so (c_k) is a Cauchy sequence. Its limit c lies in C, since C is closed, and $\|x - c\| = d$ by the continuity of the norm. If c' is another closest point, the same inequality for c and c' gives $\|c - c'\|^2 \le 0$. Now let $C = U$ be a closed subspace, and $p = P_U x$. For $u \in U$, the function $t \mapsto \|x - p - tu\|^2 = \|x - p\|^2 - 2t\langle x - p, u\rangle + t^2\|u\|^2$ is minimal at $t = 0$, so $\langle x - p, u\rangle = 0$. Conversely, if $p' \in U$ and $x - p' \perp U$, then $\|x - u\|^2 = \|x - p'\|^2 + \|p' - u\|^2$ for all $u \in U$ by Pythagoras, so p' is the closest point. Finally, $x = P_U x + (x - P_U x)$, and this decomposition is unique since $U \cap U^\perp = \{0\}$. □

For a subspace U of $\mathbb{R}^n$, this is the least squares projection of Lecture 2. The subspace must be closed: the polynomials form a subspace of $L^2[0,1]$ that is not closed, and the function e^x has the distance 0 from it (by the Weierstrass approximation theorem, Section A.15), but no closest polynomial. One can show that the random variables in L^2 that are measurable with respect to a σ-algebra $\mathcal{G}$ form a closed subspace, and the projection onto it is the conditional expectation $\mathbb{E}[Y \mid \mathcal{G}]$ (Theorem 4.15 of Lecture 4).

Theorem A.12.4 (Riesz representation)**.** *For every bounded linear functional φ on a Hilbert space H, there is exactly one $y \in H$ with $\varphi(x) = \langle x, y\rangle$ for all $x \in H$, and $\|\varphi\| = \|y\|$.*

Proof. If $\varphi = 0$, take $y = 0$. Otherwise, the kernel $U = \{x : \varphi(x) = 0\}$ is a closed subspace (the preimage of $\{0\}$ under a continuous map), and $U \neq H$. By the projection theorem, there is a $z \neq 0$ with $z \perp U$ (take $z = v - P_U v$ for some $v \notin U$), and we may scale it so that $\varphi(z) = 1$. For every x, the vector $x - \varphi(x)z$ lies in U, so it is orthogonal to z, i.e., $\langle x, z\rangle = \varphi(x)\|z\|^2$. So $y = \frac{z}{\|z\|^2}$ has the required property. If y' has it too, then $\langle x, y - y'\rangle = 0$ for all x, in particular for $x = y - y'$, so $y = y'$. Finally, $|\varphi(x)| \le \|x\|\|y\|$ by the Cauchy–Schwarz inequality, with equality for $x = y$. □

The representing vector depends on the inner product. This is why the gradient depends on the inner product (Remark 3.8 of Lecture 3): the derivative $Df(\mathbf{x})$ is a linear functional, and the gradient is its Riesz representer. The Riesz representation theorem also proves the Radon–Nikodym theorem of Lecture 4, by an argument of von Neumann.

Theorem A.12.5 (Radon–Nikodym)**.** *Let μ and ν be finite measures on a measurable space $(\Omega, \mathcal{F})$ such that $\mu(A) = 0$ implies $\nu(A) = 0$. Then there is a measurable $p \ge 0$ with $\nu(A) = \int_A p \, d\mu$ for all $A \in \mathcal{F}$.*

Sketch of proof. Let $\rho = \mu + \nu$. For $f \in L^2(\rho)$, the Cauchy–Schwarz inequality gives $|\int f \, d\nu| \le \int |f| \, d\rho \le \rho(\Omega)^{1/2}\|f\|_{L^2(\rho)}$, so $f \mapsto \int f \, d\nu$ is a bounded linear functional on $L^2(\rho)$. By the Riesz representation theorem, there is a $g \in L^2(\rho)$ with

$$\int f \, d\nu = \int fg \, d\rho = \int fg \, d\mu + \int fg \, d\nu \qquad \text{for all } f \in L^2(\rho).$$

For $f = \mathbf{1}_A$, this gives $\nu(A) = \int_A g\,d\rho$, and since $0 \le \nu(A) \le \rho(A)$ for all A, one shows that $0 \le g \le 1$ almost everywhere. Rearranging, $\int f(1-g)\,d\nu = \int fg\,d\mu$. For $f = \mathbf{1}_N$ with $N = \{g = 1\}$, the left side is 0, so $\mu(N) = 0$, and hence also $\nu(N) = 0$. Now let $A \in \mathcal{F}$, and take $f = \mathbf{1}_A(1 + g + \cdots + g^k)$, so that $f(1-g) = \mathbf{1}_A(1 - g^{k+1})$:

$$\int_A (1 - g^{k+1})\,d\nu = \int_A (g + g^2 + \cdots + g^{k+1})\,d\mu.$$

Outside the null set N, $g^{k+1} \to 0$ as $k \to \infty$, and the monotone convergence theorem (Theorem 3.8 of Lecture 4) on both sides gives $\nu(A) = \int_A \frac{g}{1-g}\,d\mu$. So $p = \frac{g}{1-g}$ is the density. □

This is Theorem 3.12 of Lecture 4 for finite measures; the σ-finite case follows by splitting Ω into countably many pieces of finite measure.

Orthonormal bases. In infinite dimensions, a basis in the sense of Lecture 1, in which every vector is a finite linear combination of basis vectors, is not useful. Instead, we allow infinite sums.

Theorem A.12.6 (Orthonormal bases)**.** *Let $e_1, e_2, \ldots$ be orthonormal vectors in a Hilbert space H, and let $x \in H$.*

1. *(Bessel's inequality) $\sum_k \langle x, e_k\rangle^2 \le \|x\|^2$.*
2. *The series $\sum_k \langle x, e_k\rangle e_k$ converges, and its sum is the orthogonal projection of x onto the closure U of* $\operatorname{span}\{e_1, e_2, \ldots\}$.
3. *If $U = H$, then $x = \sum_k \langle x, e_k\rangle e_k$ and $\|x\|^2 = \sum_k \langle x, e_k\rangle^2$ (Parseval's identity) for every $x \in H$. Then $e_1, e_2, \ldots$ is called an orthonormal basis of H.*

Proof. Let $c_k = \langle x, e_k\rangle$ and $s_n = \sum_{k\le n} c_k e_k$. Then $x - s_n \perp e_1, \ldots, e_n$, so Pythagoras gives $\|x\|^2 = \|x - s_n\|^2 + \sum_{k\le n} c_k^2$, which proves (1). For $m > n$, $\|s_m - s_n\|^2 = \sum_{k=n+1}^m c_k^2$, which tends to 0 as $n \to \infty$ by (1). So (s_n) is a Cauchy sequence, with a limit $s \in U$. By the continuity of the inner product, $\langle x - s, e_k\rangle = \lim_n \langle x - s_n, e_k\rangle = 0$ for all k, so $x - s$ is orthogonal to the span of the e_k, and, again by continuity, to its closure U. By Theorem A.12.3, $s = P_U x$. If $U = H$, then $s = x$, so $\|x - s_n\| \to 0$, and the identity $\|x\|^2 = \|x - s_n\|^2 + \sum_{k\le n} c_k^2$ gives Parseval's identity in the limit. □

So in a Hilbert space with an orthonormal basis, a vector behaves like the sequence of its coefficients: the map $x \mapsto (\langle x, e_k\rangle)_k$ is a linear bijection from H onto ℓ^2 that preserves inner products. One can show that every Hilbert space with a countable dense subset, such as $L^2[a,b]$, has an orthonormal basis, by the Gram–Schmidt process of Lecture 2 applied to a dense sequence. So all these spaces of infinite dimension are copies of ℓ^2. For example, the Gram–Schmidt process applied to $1, x, x^2, \ldots$ in $L^2[-1,1]$ gives the normalized Legendre polynomials $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\sqrt{\frac{5}{8}}\,(3x^2 - 1)$, …, which form an orthonormal basis, since the polynomials are dense in $L^2[-1,1]$.

Example A.12.7 (Fourier series)**.** The functions $\frac{1}{\sqrt{2\pi}}$, $\frac{1}{\sqrt{\pi}}\cos kx$ and $\frac{1}{\sqrt{\pi}}\sin kx$ for $k = 1, 2, \ldots$ are orthonormal in $L^2[-\pi, \pi]$, by the product formulas for sines and cosines, and one can show that they form an orthonormal basis. The expansion of a function f in this basis is its Fourier series,

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx), \qquad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos kx\,dx, \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin kx\,dx,$$

which converges in $L^2[-\pi,\pi]$. For $f(x)=x$, integration by parts gives $a_k=0$ and $b_k=\frac{2(-1)^{k+1}}{k}$, so

$$x = 2\Big(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\Big) \qquad \text{in } L^2[-\pi,\pi].$$

The coefficients with respect to the orthonormal functions $\frac{1}{\sqrt{\pi}}\sin kx$ are $\sqrt{\pi}\,b_k = \frac{2\sqrt{\pi}(-1)^{k+1}}{k}$, so Parseval's identity gives $\int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^3}{3} = \sum_k \frac{4\pi}{k^2}$, i.e.,

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

The partial sums converge to f in L^2, but not uniformly: near the jumps of the periodic extension of f at $\pm\pi$, they overshoot by about 9% of the jump, however many terms we take (the Gibbs phenomenon). The Fourier coefficients of smooth periodic functions decay quickly, so that a few of them describe the function well. Compression methods for sound and images, such as MP3 and JPEG, are based on this idea, with related bases.

A.13 Compact Operators and the Spectral Theorem

The spectral theorem of Lecture 2 says that a symmetric matrix has an orthonormal basis of eigenvectors. In infinite dimensions, a bounded symmetric operator need not have any eigenvectors: the multiplication operator $(Mf)(x) = xf(x)$ on $L^2[0,1]$ satisfies $\langle Mf, g\rangle = \langle f, Mg\rangle$, but $Mf = \lambda f$ forces $f = 0$ outside the null set $\{\lambda\}$, i.e., $f = 0$ in L^2. The spectral theorem holds for operators that are close to matrices in the following sense.

Definition A.13.1 (Compact and self-adjoint operators)**.** A bounded linear operator $K : H \to H$ on a Hilbert space is compact if for every bounded sequence (x_k) in H, the sequence (Kx_k) has a convergent subsequence. It is self-adjoint if $\langle Kx, y\rangle = \langle x, Ky\rangle$ for all $x, y \in H$, and positive semidefinite if moreover $\langle Kx, x\rangle \geq 0$ for all x.

By the Bolzano–Weierstrass theorem (Theorem A.3.1), every bounded operator whose image is finite-dimensional is compact. The identity of an infinite-dimensional Hilbert space is not compact (Remark A.3.9).

Theorem A.13.2 (Spectral theorem for compact operators)**.** *Let K be a compact self-adjoint operator on a Hilbert space H. Then there are finitely or countably many orthonormal eigenvectors $e_1, e_2, \ldots$ of K with nonzero eigenvalues $\lambda_1, \lambda_2, \ldots$, ordered so that $|\lambda_1| \geq |\lambda_2| \geq \cdots$, such that*

$$Kx = \sum_k \lambda_k \langle x, e_k\rangle e_k \qquad \textit{for all } x \in H.$$

If there are infinitely many, then $\lambda_k \to 0$, and the eigenspace of each nonzero eigenvalue is finite-dimensional. If K is positive semidefinite, all λ_k are positive.

Idea of proof. One shows that $|\langle Kx, x\rangle|$ attains its maximum on the unit ball, by compactness (this replaces the compactness of the unit sphere in $\mathbb{R}^n$), and that a maximizer is an eigenvector with the eigenvalue $\pm\|K\|$, as for the Rayleigh quotient in Lecture 2. Then one repeats this on the orthogonal complement of the eigenvectors found so far, which K maps into itself. If there were infinitely many orthonormal eigenvectors with $|\lambda_k| \geq \varepsilon > 0$, then $\|Ke_k - Ke_l\|^2 = \lambda_k^2 + \lambda_l^2 \geq 2\varepsilon^2$ for $k \neq l$, so the sequence (Ke_k) would have no convergent subsequence, although (e_k) is bounded. This shows that $\lambda_k \to 0$, and that the eigenspaces of nonzero eigenvalues are finite-dimensional. $\square$

Integral operators. Let $k : [a,b]^2 \to \mathbb{R}$ be continuous and symmetric, $k(x,y) = k(y,x)$. The integral operator

$$(Kf)(x) = \int_a^b k(x,y) f(y)\, dy$$

is the infinite-dimensional analogue of the product of a symmetric matrix with a vector, with $k(x,y)$ in the role of the entries. By Fubini's theorem, it is self-adjoint on $L^2[a,b]$, and one can show that it is compact. We call k positive semidefinite if the matrix $(k(x_i,x_j))_{i,j}$ is positive semidefinite for all points $x_1,\dots,x_n \in [a,b]$, as for the kernels of Gaussian processes in Lecture 4; then K is positive semidefinite.

Theorem A.13.3 (Mercer)**.** *Let k be continuous, symmetric and positive semidefinite on $[a,b]^2$, and let λ_k and e_k be the positive eigenvalues and the orthonormal eigenfunctions of K. Then the e_k are continuous, and*

$$k(x,y) = \sum_k \lambda_k e_k(x) e_k(y),$$

with absolute and uniform convergence on $[a,b]^2$. In particular, $\sum_k \lambda_k = \int_a^b k(x,x)\, dx$.

We do not prove Mercer's theorem. It says that $k(x,y) = \langle \boldsymbol{\phi}(x), \boldsymbol{\phi}(y) \rangle$ is the inner product in ℓ^2 of the features $\boldsymbol{\phi}(x) = (\sqrt{\lambda_k}\, e_k(x))_k$: every continuous positive semidefinite kernel is an inner product of (possibly infinitely many) features, as the kernels of the kernel trick in Lecture 2.

Example A.13.4 (The Karhunen–Loève expansion of Brownian motion)**.** The covariance function $k(s,t) = \min(s,t)$ of Brownian motion on $[0,1]$ (Proposition 14.2 of Lecture 4) is continuous, symmetric and positive semidefinite, since covariance matrices are positive semidefinite. Its integral operator is

$$(Kf)(t) = \int_0^t s f(s)\, ds + t \int_t^1 f(s)\, ds.$$

Differentiating, $(Kf)'(t) = tf(t) + \int_t^1 f(s)\, ds - tf(t) = \int_t^1 f(s)\, ds$, so $(Kf)'' = -f$, $(Kf)(0) = 0$ and $(Kf)'(1) = 0$. Hence an eigenfunction with $Ke = \lambda e$ and $\lambda > 0$ satisfies $\lambda e'' = -e$ with $e(0) = 0$ and $e'(1) = 0$. So $e(t)$ is a multiple of $\sin(t/\sqrt{\lambda})$ with $\cos(1/\sqrt{\lambda}) = 0$, which gives the normalized eigenfunctions and the eigenvalues

$$e_k(t) = \sqrt{2}\sin\big((k-\tfrac{1}{2})\pi t\big), \qquad \lambda_k = \frac{1}{(k-\frac{1}{2})^2\pi^2}, \qquad k = 1,2,\dots$$

As Mercer's theorem predicts, $\sum_k \lambda_k = \frac{4}{\pi^2}\sum_k \frac{1}{(2k-1)^2} = \frac{4}{\pi^2}\cdot\frac{\pi^2}{8} = \frac{1}{2} = \int_0^1 \min(t,t)\, dt$, where the sum over the odd numbers is $\frac{\pi^2}{6} - \frac{1}{4}\cdot\frac{\pi^2}{6}$, by Example A.12.7. Now let $Z_1, Z_2, \dots$ be i.i.d. $\mathcal{N}(0,1)$, and

$$B(t) = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, Z_k e_k(t).$$

By independence and Mercer's theorem, $\mathbb{E}[B(s)B(t)] = \sum_k \lambda_k e_k(s) e_k(t) = \min(s,t)$: B has the covariance function of Brownian motion, and one can show that the series converges and defines a Brownian motion. This is the Karhunen–Loève expansion. It writes a random function as a series with independent coefficients, ordered by their variances: it is the principal component analysis (Lecture 2) of a random function. The first term carries the fraction $\frac{\lambda_1}{1/2} = \frac{8}{\pi^2} \approx 81\%$ of the total variance $\mathbb{E}\big[\int_0^1 B(t)^2\, dt\big] = \frac{1}{2}$, and the first three terms carry 93%.

A.14 Reproducing Kernel Hilbert Spaces

Kernel methods and Gaussian processes (Lecture 4) work with functions through a kernel. The natural Hilbert spaces for them are spaces of functions in which evaluating a function at a point is continuous, unlike in L^2.

Definition A.14.1 (Reproducing kernel Hilbert space)**.** Let X be a set. A Hilbert space H of functions $f : X \to \mathbb{R}$ is a reproducing kernel Hilbert space (RKHS) if for every $x \in X$, the evaluation $f \mapsto f(x)$ is a bounded linear functional on H.

By the Riesz representation theorem, for every $x \in X$ there is a function $k_x \in H$ with

$$f(x) = \langle f, k_x \rangle \qquad \text{for all } f \in H,$$

the reproducing property. The function $k(x, y) = \langle k_x, k_y \rangle = k_y(x)$ is called the reproducing kernel of H. It is symmetric, and it is positive semidefinite: for points $x_1, \dots, x_n$ and $\mathbf{a} \in \mathbb{R}^n$,

$$\sum_{i,j} a_i a_j k(x_i, x_j) = \Big\| \sum_i a_i k_{x_i} \Big\|^2 \geq 0.$$

Conversely, by the Moore–Aronszajn theorem, which we do not prove, every symmetric positive semidefinite kernel k on X is the reproducing kernel of exactly one RKHS H_k. It is the completion of the space of the finite combinations $\sum_i a_i k_{x_i}$ with the inner product $\langle \sum_i a_i k_{x_i}, \sum_j b_j k_{y_j} \rangle = \sum_{i,j} a_i b_j k(x_i, y_j)$. In particular, $\|\sum_i a_i k_{x_i}\|^2 = \mathbf{a}^\top K \mathbf{a}$, with the kernel matrix $K_{ij} = k(x_i, x_j)$.

Example A.14.2.

1. (Linear kernel.) For $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$ on $\mathbb{R}^d$, H_k consists of the linear functions $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$, with $\langle f_{\mathbf{v}}, f_{\mathbf{w}} \rangle = \mathbf{v}^\top \mathbf{w}$: indeed, $k_{\mathbf{x}} = f_{\mathbf{x}}$, and $\langle f_{\mathbf{w}}, k_{\mathbf{x}} \rangle = \mathbf{w}^\top \mathbf{x} = f_{\mathbf{w}}(\mathbf{x})$. So the RKHS norm $\|f_{\mathbf{w}}\| = \|\mathbf{w}\|$ of a linear function is the norm of its weight vector, which ridge regression penalizes.

2. (Feature maps.) Similarly, for $k(x, y) = \boldsymbol{\phi}(x)^\top \boldsymbol{\phi}(y)$ with a feature map $\boldsymbol{\phi} : X \to \mathbb{R}^m$, H_k consists of the functions $f_{\mathbf{w}}(x) = \mathbf{w}^\top \boldsymbol{\phi}(x)$, and $\|f\|$ is the smallest norm of a $\mathbf{w}$ with $f = f_{\mathbf{w}}$. For a kernel as in Mercer's theorem, with the features $\sqrt{\lambda_k}\, e_k(x)$, H_k consists of the functions $f = \sum_k c_k e_k$ with $\|f\|^2 = \sum_k \frac{c_k^2}{\lambda_k} < \infty$. Since $\lambda_k \to 0$, the coefficients of the functions in H_k decay quickly.

3. (Smoothness.) For the kernel $\min(s, t)$ on $[0, 1]$, H_k consists of the functions $f(t) = \int_0^t g(s)\, ds$ with $g \in L^2[0, 1]$, with $\langle f_1, f_2 \rangle = \int_0^1 f_1'(s) f_2'(s)\, ds$. Indeed, $k_s(t) = \min(s, t)$ has the derivative $\mathbf{1}_{[0,s]}$, so $\langle f, k_s \rangle = \int_0^s f'(u)\, du = f(s)$. The RKHS norm $\|f\| = \|f'\|_{L^2}$ measures how much f varies. Similarly, one can show that all functions in the RKHS of the squared exponential kernel are infinitely differentiable.

The RKHS norm is a natural regularizer for learning a function from data, and the following theorem makes such learning problems finite-dimensional.

Theorem A.14.3 (Representer theorem)**.** *Let H be an RKHS on X with the kernel k, and let $\lambda > 0$. For any points $x_1, \dots, x_N$ in X and any function $L : \mathbb{R}^N \to \mathbb{R}$, every minimizer f over H of*

$$L(f(x_1), \dots, f(x_N)) + \lambda \|f\|^2$$

has the form $f = \sum_{i=1}^N \alpha_i k_{x_i}$ with $\alpha_1, \dots, \alpha_N \in \mathbb{R}$.

Proof. The span U of $k_{x_1}, \dots, k_{x_N}$ is finite-dimensional, hence closed. Write $f = u + w$ with $u \in U$ and $w \perp U$ (Theorem A.12.3). Then $f(x_i) = \langle f, k_{x_i} \rangle = \langle u, k_{x_i} \rangle = u(x_i)$, so the first term is the same for f and u, while $\|f\|^2 = \|u\|^2 + \|w\|^2$. If $w \neq 0$, then u has a smaller objective than f, so f is not a minimizer. □

So an optimization problem over an infinite-dimensional space of functions reduces to the N coefficients α_i.

Example A.14.4 (Kernel ridge regression)**.** For the squared loss, we minimize $\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \|f\|^2$. By the representer theorem, $f = \sum_j \alpha_j k_{x_j}$, so $f(x_i) = (K\boldsymbol{\alpha})_i$ and $\|f\|^2 = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha}$, and we have to minimize $\|\mathbf{y} - K\boldsymbol{\alpha}\|^2 + \lambda \boldsymbol{\alpha}^\top K \boldsymbol{\alpha}$ over $\boldsymbol{\alpha} \in \mathbb{R}^N$. This convex function has the gradient $2K\big((K + \lambda I)\boldsymbol{\alpha} - \mathbf{y}\big)$, which vanishes for $\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$. So the prediction at a new point x_* is

$$f(x_*) = \mathbf{k}_*^\top (K + \lambda I)^{-1} \mathbf{y}, \qquad \mathbf{k}_* = (k(x_1, x_*), \dots, k(x_N, x_*))^\top,$$

the posterior mean of Gaussian process regression with the noise variance $\sigma^2 = \lambda$ (Proposition 12.11 of Lecture 4). For the data of Example 12.12 of Lecture 4, with $\lambda = \sigma^2 = 0.1$, the coefficients are $\boldsymbol{\alpha} = (1.306, -0.720)$, the prediction at $x_* = 0.5$ is $f(0.5) = 0.517$, and the RKHS norm of the regression function is given by $\|f\|^2 = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha} \approx 1.084$.

A.15 Functional Analysis in Machine Learning

- **Universal approximation.** By the Weierstrass approximation theorem, which we do not prove, the polynomials are dense in $C[a, b]$: every continuous function is a uniform limit of polynomials. Since $\|f\|_{L^2} \leq \sqrt{b - a}\, \|f\|_\infty$, and one can show that the continuous functions are dense in $L^2[a, b]$, the polynomials are also dense in $L^2[a, b]$. Neural networks have the same property. Cybenko's theorem (1989) says that for a continuous function σ with $\sigma(t) \to 0$ as $t \to -\infty$ and $\sigma(t) \to 1$ as $t \to \infty$, such as the logistic sigmoid, the networks $\sum_{i=1}^m c_i \sigma(\mathbf{w}_i^\top \mathbf{x} + b_i)$ with one hidden layer are dense in $C(Q)$, the continuous functions on a box $Q \subseteq \mathbb{R}^n$ with the maximum norm. The proof is functional analysis. If the closure U of the subspace of these networks were not all of $C(Q)$, the Hahn–Banach theorem would give a nonzero bounded linear functional that vanishes on U, and by the Riesz–Markov theorem, it would be a signed measure $\mu \neq 0$ with $\int \sigma(\mathbf{w}^\top \mathbf{x} + b)\, d\mu(\mathbf{x}) = 0$ for all $\mathbf{w}$ and b. One then shows that this forces $\mu = 0$. The theorem says nothing about how many neurons are needed, or whether training finds good weights.

- **Kernel methods.** Support vector machines, kernel ridge regression and Gaussian processes work in the RKHS of a kernel, which is often infinite-dimensional. By the representer theorem, all computations only involve the $N \times N$ kernel matrix: this is the kernel trick of Lecture 2. The RKHS norm is the regularizer, and for kernels such as the squared exponential kernel, it penalizes functions that are not smooth.

- **Gaussian processes and their RKHS.** The posterior mean of a Gaussian process lies in the RKHS of its kernel, but the paths of the process are rougher than the functions in the RKHS. For Brownian motion, the RKHS consists of the functions with square-integrable derivatives (Example A.14.2), while the paths are nowhere differentiable (Section 14 of Lecture 4). In terms of the Karhunen–Loève expansion $B = \sum_k \sqrt{\lambda_k} Z_k e_k$, the RKHS norm of a path would be $\sum_k Z_k^2$, which is infinite almost surely, by the strong law of large numbers (Theorem 7.12 of Lecture 4).

- **The neural tangent kernel.** For a network $f(\mathbf{x};\boldsymbol{\theta})$, the kernel

$$\Theta(\mathbf{x},\mathbf{x}') = \nabla_{\boldsymbol{\theta}} f(\mathbf{x};\boldsymbol{\theta})^\top \nabla_{\boldsymbol{\theta}} f(\mathbf{x}';\boldsymbol{\theta})$$

  describes how a gradient step on one example changes the prediction at another input. One can show that for infinitely wide networks with a suitably scaled random initialization, this kernel stays constant during training, and that training by gradient descent on the squared loss then amounts to kernel regression with this kernel. This connects deep learning with the kernel methods of this appendix, at least in this limit.

- **Functional gradients and boosting.** Gradient boosting performs gradient descent in a space of functions: at each step, it fits a simple model, such as a small decision tree, to the negative gradient of the loss with respect to the predictions $f(\mathbf{x}_i)$, and adds a small multiple of it to the current model.

- **Neural operators.** Some learning problems map functions to functions, for example the initial condition of a differential equation to the solution at a later time. Neural operators learn such maps between infinite-dimensional spaces, for example by acting on the Fourier coefficients of their inputs (Example A.12.7).

Key Takeaways

- Open sets, closed sets, convergence and continuity only depend on the notion of nearness. A map is continuous if and only if preimages of open sets are open; so sublevel sets $\{f \le c\}$ and level sets of continuous functions are closed. All norms on $\mathbb{R}^n$ define the same open sets.

- A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded (Heine–Borel). A continuous function on a nonempty compact set attains its minimum and maximum, and a continuous coercive function on $\mathbb{R}^n$ has a global minimum; regularization makes losses coercive.

- Continuous maps preserve connectedness; this gives the intermediate value theorem, and shows, e.g., that $\mathrm{GL}(n,\mathbb{R})$ has two pieces and that a circle is not homeomorphic to an interval.

- A contraction of a closed set has a unique fixed point, found by iteration (Banach). This proves the inverse function theorem: a map with an invertible Jacobian is locally invertible, with $D(\mathbf{f}^{-1}) = (D\mathbf{f})^{-1}$. Residual layers $\mathbf{x} + \mathbf{g}(\mathbf{x})$ with a contractive $\mathbf{g}$ are invertible.

- The implicit function theorem solves $F(\mathbf{x},\mathbf{y}) = \mathbf{0}$ locally for $\mathbf{y} = \mathbf{g}(\mathbf{x})$ when $D_{\mathbf{y}}F$ is invertible, with $D\mathbf{g} = -D_{\mathbf{y}}F^{-1}D_{\mathbf{x}}F$; this is implicit differentiation, also used for implicit layers and hyperparameter optimization.

- The level set of a regular value of $F : \mathbb{R}^n \to \mathbb{R}^m$ is an $(n-m)$-dimensional submanifold, which is locally a graph. Its tangent space is $\operatorname{null} DF(\mathbf{p})$, and its normal space is spanned by the gradients $\nabla F_i(\mathbf{p})$. Examples: spheres, $\mathrm{SL}(n)$ (dimension n^2-1, tangent space at I: traceless matrices), $\mathrm{O}(n)$ (dimension $\frac{n(n-1)}{2}$, tangent space at I: skew-symmetric matrices).

- Lagrange multipliers say that ∇f is normal to the constraint manifold at a constrained extremum. The Riemannian gradient is the projection of ∇f onto the tangent space; Riemannian gradient descent alternates such steps with retractions.

- By Sard's theorem, almost all values of a smooth map are regular. A critical point is nondegenerate if its Hessian is invertible, and its index is the number of negative eigenvalues. Nondegenerate critical points are isolated, and by the Morse lemma f is exactly a sum of $\pm y_i^2$ near them. Almost every linear perturbation of a function is a Morse function.

- Sublevel sets change their topology only at critical values: a critical point of index k attaches a k-cell. On a compact manifold, $\chi(M) = \sum_k (-1)^k c_k$.

- A Banach space is a complete normed space, such as $C[a, b]$ with the maximum norm, ℓ^p and L^p. The Banach fixed-point theorem holds in it; with a weighted maximum norm, it proves the existence and uniqueness of solutions of ODEs (Picard–Lindelöf). A linear map is continuous if and only if it is bounded; integration is bounded, differentiation is not.

- In a Hilbert space, every closed convex set has a unique closest point (projection theorem), every bounded linear functional is an inner product with a fixed vector (Riesz), and an orthonormal basis gives the expansion $x = \sum_k \langle x, e_k \rangle e_k$ with Parseval's identity, such as Fourier series. Conditional expectations are projections in L^2, and the Riesz representation theorem proves the Radon–Nikodym theorem.

- A compact self-adjoint operator has orthonormal eigenvectors with eigenvalues that tend to 0, and it is diagonal in them. For a continuous positive semidefinite kernel, Mercer's theorem gives $k(x, y) = \sum_k \lambda_k e_k(x) e_k(y)$, and the Karhunen–Loève expansion of a random function, such as Brownian motion, is its principal component analysis.

- In an RKHS, evaluation is continuous, $f(x) = \langle f, k_x \rangle$, and every positive semidefinite kernel has one. By the representer theorem, regularized learning in an RKHS reduces to N coefficients; kernel ridge regression gives $\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$, the posterior mean of a Gaussian process. Networks with one hidden layer are dense in the continuous functions (Cybenko).

Exercises

1. Which of the following subsets of $\mathbb{R}^2$ are open, closed, both or neither? Find their interior, closure and boundary: (a) $\{x > 0, y > 0\}$; (b) $\{x^2 + y^2 \leq 1\}$; (c) $\{(x, 0) : 0 \leq x < 1\}$; (d) $\mathbb{Q}^2$, the points with rational coordinates.

2. Show that the union of infinitely many closed sets need not be closed. Show that a set $U \subseteq X$ is open if and only if it is a union of open balls.

3. (a) Show that the set of $m \times n$ matrices of rank at most r is closed in $\mathbb{R}^{m \times n}$. (Hint: the rank is at most r if and only if all $(r + 1) \times (r + 1)$ minors are 0.)

   (b) Show that every square matrix A is a limit of invertible matrices. (Hint: $\det(A + tI)$ is a polynomial in t.) So $\mathrm{GL}(n, \mathbb{R})$ is open and dense in $\mathbb{R}^{n \times n}$.

4. Which of the following sets are compact? (a) The closed unit ball of the ℓ_1 norm in $\mathbb{R}^n$; (b) $\{(x, y) : xy = 1\}$; (c) $\mathrm{SO}(n)$; (d) $\mathrm{GL}(n, \mathbb{R})$; (e) the ellipsoid $\{\mathbf{x} : \mathbf{x}^\top A \mathbf{x} \leq 1\}$ for a symmetric positive definite A; (f) the same set for $A = \mathrm{diag}(1, -1)$.

5. Show that the regularized logistic regression loss $\frac{1}{N} \sum_i \log(1 + e^{-y_i \mathbf{w}^\top \mathbf{x}_i}) + \lambda \|\mathbf{w}\|^2$ with $\lambda > 0$ has a global minimum, and that it is unique. (Use Proposition A.3.6 and Lecture 3.)

6. Let $A \in \mathbb{R}^{n\times n}$. Show that $N(\mathbf{x}) = \|A\mathbf{x}\|$ is a norm if and only if A is invertible, and that the best constants in $c\|\mathbf{x}\| \le N(\mathbf{x}) \le C\|\mathbf{x}\|$ are the smallest and the largest singular value of A (Lecture 2).

7. (a) Show that the unit circle S^1 is path-connected, and that the hyperbola $\{x^2 - y^2 = 1\}$ is not connected.
   (b) Show that $\mathrm{SO}(2)$ is path-connected, and that $\mathrm{O}(2)$ is not connected.
   (c) Show that no continuous injective map from S^1 to $\mathbb{R}$ exists. (Hint: intermediate value theorem.)

8. Use the iteration $x_{k+1} = \cos x_k$ with $x_0 = 1$ to compute the solution of $\cos x = x$ to six decimal places. How many steps does the a priori bound of Theorem A.5.3 require, and how many do you actually need?

9. Show that $f(x) = x + \frac{1}{2}\tanh(x)$ is a homeomorphism of $\mathbb{R}$, and compute $f^{-1}(1)$ with the iteration of Corollary A.5.5, starting at $x_0 = 0$. Is f a diffeomorphism?

10. At which points of the folium $x^3 + y^3 = 3xy$ can the curve not be written locally as a graph $y = g(x)$? Compute $g'(x)$ at the point $(\frac{3}{2}, \frac{3}{2})$.

11. Let $F(x, y, z) = x^2 + y^2 - z^2$. Find all critical points and critical values of F. For which c is $F^{-1}(c)$ a submanifold? Describe the level sets for $c > 0$, $c = 0$ and $c < 0$; which of them are connected?

12. (a) Find the tangent space of the sphere $x^2+y^2+z^2 = 9$ at $(1, 2, 2)$ as null DF, and the tangent plane $(1, 2, 2) + T_{(1,2,2)}M$ as an equation.
   (b) Show that $T_Q\mathrm{O}(n) = \{QK : K^\top = -K\}$ and $T_I\mathrm{SL}(n, \mathbb{R}) = \{H : \operatorname{tr} H = 0\}$ have the dimensions predicted by Example A.7.3.

13. (Orthogonal Procrustes problem.) Given $A \in \mathbb{R}^{n\times n}$ with SVD $A = U\Sigma V^\top$, show that $\operatorname{tr}(Q^\top A)$ is maximal over $Q \in \mathrm{O}(n)$ for $Q = UV^\top$. (Hint: $\operatorname{tr}(Q^\top A) = \operatorname{tr}(\Sigma Z)$ with the orthogonal matrix $Z = V^\top Q^\top U$, whose entries are at most 1 in absolute value.) This is how one aligns two point clouds by a rotation or reflection.

14. (a) Find all critical points of the monkey saddle $f(x, y) = x^3 - 3xy^2$. Show that the origin is a degenerate critical point. In polar coordinates, $f = r^3 \cos 3\varphi$; describe the regions where $f > 0$ and $f < 0$.
   (b) For which $a \in \mathbb{R}$ is $x^3 - ax$ a Morse function? Find its critical points and their indices.
   (c) Find the critical points and their indices of $f(x, y) = x^4 - 2x^2 + y^2 + \varepsilon x$ for a small $\varepsilon > 0$, and describe how the sublevel sets change as c increases.

15. Let $A = \operatorname{diag}(3, 2, 1)$ and $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on the sphere S^2. Find all critical points of f and their indices, and check the Morse relation $\chi(S^2) = 2$. Repeat for $A = \operatorname{diag}(4, 3, 2, 1)$ on S^3, whose Euler characteristic is 0.

16. (a) Show that $f_n(x) = \sqrt{x^2 + \frac{1}{n}}$ converges to $|x|$ uniformly on $[-1, 1]$. Conclude that the continuously differentiable functions do not form a closed subset of $C[-1, 1]$.
   (b) Show that $f_n(x) = x^n$ is a Cauchy sequence in $L^2[0, 1]$, but not in $C[0, 1]$ with the maximum norm.

17. (A Neumann series.) Let V be the Volterra operator of Example A.11.5. Show that the equation $f = 1 + Vf$, i.e., $f(x) = 1 + \int_0^x f(t)\,dt$, has exactly one solution in $C[0,1]$ (Hint: such an f solves a differential equation), and that the iteration $f_0 = 0$, $f_{k+1} = 1 + Vf_k$ gives the partial sums of the series $\sum_{j\geq 0} V^j 1$, whose terms have the norms $\frac{1}{j!}$. Which function is the solution?

18. (The best quadratic approximation.) Use the normalized Legendre polynomials of Section A.12 to find the polynomial p of degree at most 2 that minimizes $\int_{-1}^{1}(|x| - p(x))^2\,dx$, and compute the minimal value.

19. Compute the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$, and use Parseval's identity to show that $\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$. Why do the partial sums converge uniformly here, unlike for $f(x) = x$? (Hint: compare the decay of the coefficients.)

20. (Riesz representers.)

    (a) Let A be symmetric positive definite, and give $\mathbb{R}^n$ the inner product $\langle \mathbf{x}, \mathbf{y}\rangle_A = \mathbf{x}^\top A\mathbf{y}$. Find the vector that represents the functional $\mathbf{x} \mapsto \mathbf{c}^\top \mathbf{x}$.

    (b) Find the function in $L^2[0,1]$ that represents $f \mapsto \int_0^{1/2} f(x)\,dx$.

    (c) Show that $f \mapsto f(\frac{1}{2})$ is not bounded on $C[0,1]$ with the norm of $L^2[0,1]$. (Hint: narrow tents.)

21. Check by integrating that $e_1(t) = \sqrt{2}\sin\frac{\pi t}{2}$ is an eigenfunction of the integral operator of Example A.13.4 with the eigenvalue $\frac{4}{\pi^2}$, and that $\|e_1\|_{L^2} = 1$.

22. (Minimal-norm interpolation.) Let $0 < t_1 < \cdots < t_N \leq 1$ and $y_1, \ldots, y_N \in \mathbb{R}$. In the RKHS of the kernel $\min(s,t)$ on $[0,1]$ (Example A.14.2), find the function f with the smallest norm such that $f(t_i) = y_i$ for all i. (Hint: as in the proof of the representer theorem, $f = \sum_i \alpha_i k_{t_i}$.) Show that f is the piecewise linear function through the points $(0,0), (t_1, y_1), \ldots, (t_N, y_N)$, constant on $[t_N, 1]$.

23. For the linear kernel $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$, show that the prediction $\mathbf{k}_*^\top (K + \lambda I)^{-1}\mathbf{y}$ of kernel ridge regression is $\mathbf{x}_*^\top \hat{\mathbf{w}}$ with the ridge regression solution $\hat{\mathbf{w}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$, where X has the rows $\mathbf{x}_i^\top$. (Compare with the exercise on Gaussian processes and Bayesian linear regression in Lecture 4.)

Programming Exercises

As in Lecture 3, use NumPy (and Matplotlib for plots), and start each file with

```
import numpy as np
import matplotlib.pyplot as plt
rng = np.random.default_rng(0)
```

P1. Inverting a residual layer. Let `W = rng.normal(size=(5, 5))` and $\mathbf{g}(\mathbf{x}) = c\tanh(W\mathbf{x})$ with $c = 0.9/\|W\|_2$.

- Show that $\mathbf{g}$ is Lipschitz continuous with constant 0.9 (use $|\tanh'| \leq 1$ and Lemma A.5.1).
- For a random $\mathbf{x}$, compute $\mathbf{y} = \mathbf{x} + \mathbf{g}(\mathbf{x})$, and recover $\mathbf{x}$ with the iteration of Corollary A.5.5. Plot the error against the number of iterations on a logarithmic scale, together with the bound of Theorem A.5.3.

- Repeat with $c = 1.5/\|W\|_2$. Does the iteration still converge? Is $\mathbf{x} \mapsto \mathbf{x} + \mathbf{g}(\mathbf{x})$ still injective? (Try to find two points with the same image.)

P2. Riemannian gradient ascent on the sphere. For a random symmetric $A = B + B^\top$ with `B = rng.normal(size=(6, 6))`, maximize $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on S^5 with the Riemannian gradient and the normalization retraction (Section A.8).

- Start from 100 random points, and check that you always reach $\pm\mathbf{q}_1$ (compare with `np.linalg.eigh`). Plot $f(\mathbf{p}_k)$ for a few runs.
- Start exactly at an eigenvector $\mathbf{q}_i$ with $i > 1$. What happens? Now add a perturbation of size 10^{-8}. How many steps does it take to escape from the saddle point, and how does this depend on $\lambda_1 - \lambda_i$?

P3. Orthogonal Procrustes. Generate a point cloud `X = rng.normal(size=(50, 3))`, a random orthogonal matrix Q_0 (from `np.linalg.qr`), and $Y = XQ_0^\top + 0.01\cdot\text{noise}$. Recover Q_0 by maximizing $\operatorname{tr}(Q^\top Y^\top X)$ over $\mathrm{O}(3)$ with the SVD formula of Exercise 13. Then solve the same problem with Riemannian gradient ascent on $\mathrm{O}(3)$: project $G = \nabla_Q$ onto $T_Q\mathrm{O}(3)$ by $Q\,\frac{1}{2}(Q^\top G - G^\top Q)$, take a step, and retract with the Q factor of a QR decomposition (fixing the signs so that the diagonal of R is positive). Compare the results.

P4. Counting components of sublevel sets. For $f(x, y) = x^4 - 2x^2 + y^2 + 0.3x$, evaluate f on a fine grid over $[-2, 2]^2$, and for 50 values of c, count the connected components of the grid points with $f \le c$ (write a breadth-first search over neighboring grid points). Plot the number of components against c, and compare the values of c where it changes with the critical values of f, which you can find with Newton's method applied to $\nabla f = \mathbf{0}$.

P5. Picard iteration. Consider the logistic equation $x' = x(1 - x)$ with $x(0) = 0.2$, whose solution is $x(t) = \frac{1}{1+4e^{-t}}$.

- On a grid of 1001 points in $[0, 5]$, implement the Picard iteration $x_{k+1}(t) = 0.2 + \int_0^t x_k(s)(1 - x_k(s))\,ds$ of Theorem A.11.4, starting from $x_0(t) = 0.2$ and computing the integrals with the cumulative trapezoidal rule. Plot the iterates $x_0, \dots, x_8$ together with the solution.
- Plot the maximum error of x_k against k on a logarithmic scale, on $[0, 1]$ and on $[0, 5]$. Where does the iteration converge first, and why?

P6. Fourier series and the Gibbs phenomenon.

- Plot the partial sums of the Fourier series of $f(x) = x$ on $[-\pi, \pi]$ with 5, 20 and 100 terms (Example A.12.7), and compute their errors in the L^2 norm and in the maximum norm. Which error tends to 0?
- For $N = 50, 100, 200, 400, 800$ terms, compute the maximum of the partial sum on a fine grid near π, and the overshoot above π as a fraction of the jump 2π. Compare with the limit ≈ 0.0895.
- Check Parseval's identity numerically, and compare the partial sums of $\sum_k \frac{1}{k^2}$ with $\frac{\pi^2}{6}$. How fast do they converge?

P7. The Karhunen–Loève expansion. Let $t_i = \frac{i}{n}$ for $i = 1, \dots, n$ with $n = 500$.

- The matrix $\frac{1}{n}\big(\min(t_i, t_j)\big)_{i,j}$ approximates the integral operator of Example A.13.4 (why?). Compute its largest eigenvalues with `np.linalg.eigh`, and compare them with $\frac{1}{(k-1/2)^2\pi^2}$. Compare the eigenvectors, scaled by $\sqrt{n}$, with $\sqrt{2}\sin\big((k-\frac{1}{2})\pi t\big)$.
- Sample Brownian paths from the expansion truncated after 5, 20 and 200 terms, and compare them with paths simulated as random walks (Lecture 4). Check $\mathbb{E}[B(0.5)^2] = 0.5$ for the truncated expansions with 10^4 samples.

P8. Kernel ridge regression.

- Implement kernel ridge regression with the squared exponential kernel (Example A.14.4), and check that it gives the posterior mean of your Gaussian process regression in the programming exercises of Lecture 4, with $\lambda = \sigma^2$. Plot the fits for $\lambda = 10^{-4}$, 10^{-2} and 1.
- For the polynomial kernel $k(x, y) = (1 + xy)^2$ on $\mathbb{R}$, show that $k(x, y) = \boldsymbol{\phi}(x)^\top \boldsymbol{\phi}(y)$ with $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$, and check numerically that kernel ridge regression gives the same predictions as ridge regression with these features. Compare the running times of the two methods for $N = 10^2, 10^3, 10^4$ data points.